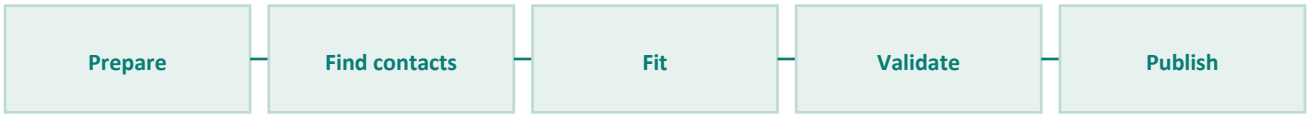
CYRUS SCATTER / RESEARCH LIBRARY

Vegetation interaction

A consolidated decision library for rooted vegetation around fixed obstacles

24 October 2026 | Research and proposed design

Prepare the source once. Keep accepted roots planted. Fit relevant contacts together. Validate the visible mesh and publish a complete result.



09 core documents / 14 preserved source files / 68 proposed validation cases

This handbook consolidates the supplied research and records corrections, proposed workflows and experiment priorities. It does not report an implemented vegetation feature or executed simulation results.

The editable Markdown, checklists, source manifest and original files accompany this PDF. Start with README.md for the complete document register.

Contents

01 Findings and recommendation 3

02 Artist workflow 5

03 Source preparation 8

04 Collision and solvers 11

05 Architecture and output 17

06 Houdini and tyFlow workflow comparison 22

07 Validation and prototype plan 24

08 Engineering guidelines 27

09 Evidence, corrections and open questions 30

Asset audit worksheet 36

Validation matrix 38

Solver decision experiments 42

Corrected recipe planning matrix 45

Original research register 47

Primary reference register 49

Read Chapters 01, 02 and 09 for product decisions. Use Chapters 03-05 and 08 for engineering, Chapter 06 for vendor workflows, and Chapter 07 with the appendices for experiments.

01 Findings and recommendation

Status: research synthesis and proposed scope. Source families: R01-R04 engineering studies, R08/R11 preparation studies, R09/R10 practical workflows and R12-R14 solver-reuse research. The final supplied study is included; all fourteen original files are preserved. See the source register and corrections.

Feasibility

The original idea is technically plausible: keep grass planted beside a stone and bend the nearby vegetation into a more natural arrangement. A useful first product is a static geometric result obtained through a bounded fitting operation. The artist can press Update while an internal numerical method performs several steps. Continuous timeline simulation is not a prerequisite for that interaction.

Houdini documents both a static curve collision operation and a Vellum mode that rolls inputs forward internally for each evaluated frame. These establish relevant building blocks, not a proven Cyrus implementation or a guarantee for arbitrary plant meshes. Guide Collide With VDB, Vellum Solver.

The hardest uncertainties are source preparation, reliable initial-intersection recovery, and delivery of differently bent geometry through Max's viewport/render/Bake paths. A convincing guide-only demonstration would leave all three partially unanswered.

R11 strengthens the representation evidence with two supplied public meshes, synthetic counterexamples and reported arithmetic checks. Independent topology recounts agree with its public-asset counts. This supports conservative preparation and explicit identity mapping; it does not establish automatic blade recognition, successful contact fitting or production asset coverage.

R13 adds a preserved C++ reproduction package and reported Linux headless diagnostics. It demonstrates, within that recorded build, a stock rod crossing an obstacle, ambiguous mesh occupancy, tiny-scale query misses and residual penetration after a rigid body sleeps. We inspected its harness/results and matched 48 cited file hashes to the local pinned dependency; we did not repeat the engine runs. This is stronger evidence for the identified gaps, not a complete fitted-grass or Max result. See the evidence review.

R14 adds a different CPU build profile, seven reported native records and nine analytic fixtures with fifteen stored checks. Its three-point rod still crosses a sphere at both 5 and 50 soft iterations; its free interior rod can be ejected with continuing velocity rather than settle into an acceptable shape. It narrows the public callback boundary and strengthens immutable query-generation ownership. The R14 review preserves these findings without treating them as our reproduced runs or as a completed fitter.

Separate the requested behaviors

Behavior	What changes	Recommended scope
Rooted vegetation fit	Shape changes; accepted root placement stays fixed	First bounded research prototype
Rigid settle	Object translation and rotation change; source shape stays fixed	Separate existing Physics Lab context and future integration decision
Deformable leaf drape	Unrooted surface bends, folds and moves	Later surface-solver investigation
Plant-to-plant interaction	Contact couples several deformable plants	Later opt-in feature with a separate cost budget

A leaf modeled as one rigid body cannot fold over a curb. A single clump guide can provide a broad push but cannot promise individual blade contact on opposite sides of a stone. A solver choice should follow the intended representation and visible result.

First product hypothesis

Use one prepared grass source with explicit roots, ordered guides, base frames and a verified guide-to-mesh binding. Begin with a small set of fixed solid primitive/convex/compound obstacles, plus explicitly declared terrain surfaces where qualified. Arbitrary mesh-solid containment needs a separate gate. Fit only affected accepted placements after ordinary placement acceptance has completed. Keep untouched placements on the existing shared-source path and attach optional deformation state to affected identities.

A focused deterministic fitter with explicit whole-segment/envelope contact is the leading hypothesis. Compare a direct triangle BVH with Jolt shape queries as its contact backend, and retain a small rod/constraint mechanics comparator. R12 and our pinned-source checks identify sampled-vertex contact and root-mass coupling in the stock CPU rod path; existing rigid-body integration does not remove those gaps. Select the method from contact quality, failure behavior, total update cost and output fidelity, not dependency reuse alone.

R13 favors selective Jolt query reuse, conditional on a qualified geometry adapter. Keep the BVH comparison: its reported tiny-scale and containment failures mean query availability alone is insufficient. Compare a fitter and rods against the same final validator, including the contact adapter each actually needs. The reported one-segment hard-clamped rod is a detection diagnostic; it cannot fairly rank feasible routing quality or speed.

The final decision remains a narrow experiment, not a production solver adoption: test a short prepared guide with enough root freedom and length for a known feasible route around one stone. Implement the missing contact response, then test visible width and two obstacles. Pair every intended rejection with a feasible control that must succeed; an implementation that rejects every plant does not meet the requested feature. Keep the authored viewport/render/Bake feasibility experiment early.

Start with artist-facing controls for obstacle selection, clearance, flexibility, root attachment and Update. Add complexity when the fixtures demonstrate a need. No arbitrary-asset automation, full ecosystem simulation, GPU requirement or universal renderer claim is justified by the present evidence.

What “closest objects” should mean

Use proximity to find a bounded neighborhood, then test the actual geometry. The nearest object origin or nearest root is insufficient: a long blade can touch a farther stone, and two stones can constrain the same blade simultaneously. Candidate search must account for plant reach, obstacle bounds and the permitted deformation envelope.

The solver needs all relevant contacts in that neighborhood. Choosing only the closest stone or averaging opposing normals can produce an apparently smooth shape that still penetrates another obstacle.

Research confidence

Conclusion	Confidence and limitation
Prepared rooted grass can be represented by constrained guides or ribbons	Strong technical basis; actual source-library suitability is unmeasured
A static artist workflow can contain internal relaxation	Documented vendor precedent; Cyrus scheduling and costs remain unimplemented

Conclusion	Confidence and limitation
Position, tangent and cross-section orientation need distinct treatment	Geometric requirement; UI defaults and material calibration remain proposed
Differently bent instances may require distinct visible geometry	Geometric consequence; exact allocation and reuse depend on the output design
A local fit can preserve accepted placements	Proposed feature invariant; upstream placement changes still follow existing contracts
Research estimates predict production performance	Unsupported; arithmetic scenarios are useful for budgeting only

Decision boundary

Proceed to a later, explicitly authorized end-to-end prototype if the user chooses implementation. First demonstrate one bent blade beside one fixed stone, with matching Max 2027 viewport, selected final renderer and Bake output. In parallel, audit a small representative asset set. Expand to clumps, two obstacles and scale only after those gates pass.

A tiny native segment-crossing/root probe can run alongside the authored output experiment. This reconciles R12's early solver falsification with the existing early output gate: both uncertainties should be tested before building a substantial solver. The decision experiments remain proposed work.

The current request authorizes this research library. It does not start that prototype or alter the active improvement-program ledger.

02 Artist workflow

Status: proposed behavior for discussion and later qualification. Existing Manual/Live and publication contracts continue to apply. Names below are working labels, not delivered controls.

First use

1. Select a Scatter layer or an explicitly supported subset of its accepted placements.

2. Choose Rooted fit and a prepared source preset. Review root positions, base frames and blade/guide associations in an overlay.

3. Add stones or other supported geometry to an explicit obstacle set. Preview the actual collision representation and clearance envelope.

4. Adjust clearance and flexibility; choose a fixed base frame or soft angular attachment. Root position remains planted in rooted mode.

5. Press Update. Review affected, unchanged and unresolved placements, then keep the completed result or revise the request.

Source preparation should be reusable. Moving a stone should generally require another fit, not another mesh binding. A source topology/reference change can require preparation again.

Essential controls

Control	Artist meaning	Required clarification
Obstacles	Geometry that this vegetation must avoid	Independent from placement-spacing scopes; ground participates only when included
Clearance	Desired visible separation in scene units	Show the full effective envelope; distinguish numerical tolerance from desired spacing
Flexibility	How strongly the plant resists bending	Start from tested source presets; avoid implying measured botanical calibration
Root position	Attachment location	Fixed in this mode; conflicts are reported
Base attachment	How the blade can turn above its root	Position-only hinge, soft angular attachment and fixed frame are different choices
Preserve source shape	Preference for the authored rest shape	A completed fit does not silently become a new rest shape
Quality/work limit	Allowed solve and validation effort	A budget limit can produce an unresolved result
Update / Cancel / Reset / Bake	Recompute, stop, restore authored shape, or create output	Define Undo and persistence before implementation

Rigid Settle needs gravity, hulls, friction and a declared start state. Drape needs surface topology, thickness, stretch and optional self-contact. Their controls should be presented with their respective modes.

After moving a stone

Manual mode marks relevant vegetation Pending and retains the last complete publication until explicit Update. That retained geometry may intersect the newly moved stone; its status must show that it is stale. Update includes the stone's old affected region, new affected region and previously dependent plants, so vegetation released from contact can return toward its rest shape.

Live mode schedules bounded work for relevant dependencies. Interactive edits can be coalesced, but a relevant animated obstacle or source cannot be ignored under a blanket “after drag release only” rule. Navigation, selection and unrelated animation must not refit static vegetation or upload unchanged buffers. These follow the existing F04 evaluation contract.

A proposed Freeze at sampled time option would need its own explicit time/dependency contract. It must not silently redefine existing Live behavior. Bake is a distinct output action with deliberate source and dependency retention rules.

Result states and recovery

State	Meaning	Useful response
Preparation needed	Roots, correspondence or geometry are unsuitable	Select the source and show the missing preparation

State	Meaning	Useful response
Pending	Relevant inputs changed	Keep prior output visibly marked stale; offer Update
Root conflict	A fixed root or protected base lies inside the obstacle envelope	Show the root and conflicting obstacle
Ambiguous obstacle	Solid occupancy or surface sidedness cannot be established	Highlight the obstacle and request a supported solid proxy or explicit sheet policy
Unsupported scale / transform	The input falls outside the qualified numerical or shape profile	Show the affected source and supported profile; do not silently change geometry or clearance
Unresolved contact	Requested clearance was not achieved	Show all participating contacts and measured residuals
Envelope too conservative	The selected circular proxy blocks a space that may admit an oriented blade	Offer a qualified orientation-aware profile or a revised request; do not call every possible blade route infeasible
Excessive deformation	Length, frame or shape limit was exceeded	Inspect binding, representation and requested path
Work limit reached	The attempt exhausted its declared budget	Report work used; do not label the result converged
Cancelled / stale successor	Work was stopped or its inputs changed	Retire staged work and keep the prior complete publication
Output unavailable	The fitted shape cannot be delivered faithfully	Keep the existing complete result and identify the failed output path

Failure to find a local solution is not proof that no possible three-dimensional shape exists. Reserve a proven infeasible label for a specific contradiction, such as a protected root inside a forbidden volume under the declared rules.

Partial acceptance is a product choice

The safe default for an unaccepted attempt is to retain the previous complete result. A useful later option could publish a complete generation containing successful fits and explicitly flagged exceptions. Such an option needs a documented choice of compatible previous shape or rest shape for every failed placement. It must preserve the accepted population, label stale exceptions and allow selection of their IDs.

This is different from exposing unfinished worker buffers. A partial-acceptance policy must still publish atomically and must never silently delete plants, move roots or lower requested clearance.

Reset, save and Undo

Reset to source removes the fit while preserving authored placements. Update starts from the declared source/rest state; use of a prior result as an optimization must preserve the declared repeatability policy. Continue simulation from current state would be a separate future operation.

Describe whether the selected mode validates only the completed static shape or also promises a collision-free path while the plant moves. Whole-segment clearance at one pose does not establish continuous clearance during deformation. The first static mode should make only the supported completion promise.

Save/load must distinguish authored preparation, current fit, stale fit and unavailable dependencies. Undo/Redo must restore the corresponding settings and completed output together, within a measured memory policy. File

loading, viewport reading and rendering should consume a completed publication rather than silently solve a new arrangement.

03 Source preparation

Status: proposed asset contract and audit method, with scoped offline representation evidence. R08 supplies the earlier preparation study; R11 adds a concrete framed-guide mapping, two public assets and supplied arithmetic/topology results. No production asset library, automatic preparation system or Max delivery was qualified. See the R11 evidence review.

Choose the representation deliberately

Tier	Suitable use	Preparation and limitation
A: whole-clump guide/cage	Broad deflection for middle or distant views	Few controls; cannot guarantee independent blade paths around a stone
B: blade/card guides or ribbons	Close grass, narrow gaps and multiple obstacles	Explicit membership, roots and frames; strongest initial product candidate for prepared assets
C: authored advanced driver	Branches, wide leaves, complex scans and existing rigs	Specialized cage, skeleton or surface binding; separate qualification

The first source should be intentionally prepared. Automatic discovery can be investigated after an audit identifies which geometry families actually dominate the intended library.

Preparation record

Store one immutable record per compatible source revision, with the following information:

- Source identity, geometry/topology signature, reference pose, units and local coordinate system.
- Component/blade membership, declared root positions or patches, root tangents and width/frame directions.
- Ordered rest guides or driver surface, rest lengths, sample spacing and a declared frame convention.
- Mesh-to-driver correspondence, weights or local coordinates, unbound vertices and cross-piece restrictions.
- Collision envelope policy, supported scale/transform range and any protected attachment region.
- UV/map channels, material IDs, smoothing and explicit-normal policy, plus preparation version and diagnostics.
- Separate render-corner, logical-position and semantic-part identities; verified duplicate correspondence and sidedness classification.
- Fixed rest binding coordinates, complete residual offsets, rest directors and any unwrapped twist; shared-base and parent/child attachment policies.

This is a proposed data contract, not a new serialized schema. Existing recipe and scene formats require a separate design decision before extension.

Automatic preparation limits

Disconnected mesh components can be useful candidates for separate blades. A connected component is not necessarily a botanical blade: joined roots, welded cards, branching stems and scans can violate that assumption. Conversely, UV seams or split shading normals can divide vertex indices without dividing the physical surface.

An optional virtual weld must use explicit tolerances, preserve analysis provenance and show ambiguous merges. A root-cut analysis graph may help separate joined blades while leaving the original mesh intact. Neither method should destructively detach or rewrite the artist's source.

R11 makes the distinction concrete. An independent recount of the assets embedded in its evidence archive matched these results:

Public source	OBJ position vertices / components	Expanded vertices / components	Triangles
Kenney grass	84 / 9	264 / 66	132
Kenney grass_leafs	30 / 6	84 / 24	36

Expansion here uses each distinct position/UV/normal index tuple; the supplied GLB counts agree. None of these component counts establishes botanical blade identity. Both GLBs are opaque materials, so these are not tests of alpha-masked foliage. The recount inspected archive data without running its bundled scripts; provenance and exact scope are recorded in the review receipt.

Each OBJ triangle also has an opposite-winding mate using the same position IDs. Such explicit two-sided construction can inflate raw face-incidence counts. A reversible analysis-only representative can simplify that graph while preserving every original render face and channel. Do not classify the source as corrupt, or delete its back faces, merely from a raw nonmanifold-edge count. Deliberately independent coincident blades must remain separate logical positions.

A shared vertex has one output position. Two independent blade drivers cannot assign it incompatible motion. Choose an explicit common base, pinned transition, reviewed blend or a separately authorized topology change. For attached leaves, evaluate the attachment in the parent's current frame; a child attachment is not a pin to its old world position.

Simple ribbons can support a root-to-tip traversal if width, branching and ordering checks pass. A height threshold is acceptable for a controlled upright fixture, not a general terrain or curved-source root finder. Show suggested roots and frames before treating them as accepted preparation.

Opacity cards contain geometry and a shading silhouette. One textured card showing several grass blades can bend as a card; it does not provide independent geometry for every pictured blade. Alpha-pixel collision and reconstruction are outside the first scope.

Opaque renderer proxies require a verified path to full canonical geometry and attributes. A coarse preview representation is insufficient for a source contract or final contact validation.

Frames and mesh decoding

A centerline supplies position and tangent. It needs a cross-section direction to determine ribbon orientation. Two fixed collinear root samples constrain a tangent but leave axial roll free. A noncollinear base patch can define a geometric frame. For a soft base, the root position can remain fixed while a separate angular constraint permits bending.

Use one explicitly tested decoder first. A framed single-guide mapping or a small ribbon driver is easier to verify than a universal skin/cage abstraction. Interpolation must not blend nearby independent blades merely because their vertices are close.

For vendor comparisons, SideFX warns that ordinary Point Deform is unsuitable for wire models; a piece attribute does not remove that limitation. A curve-aware framed decoder or an actual surface/volume cage needs its own correspondence checks. Point Deform, Wire Deform.

Dual-quaternion skinning can be a useful option, but a fixed four-influence DQ design is not established as the right universal decoder. Matching an imported Max Skin result also requires its deformation mode and per-vertex DQ blend behavior, not only bone weights. ISkinContextData.

Framed-offset reference candidate from R11

R11 proposes a specific mapping for already prepared bindings. For each logical vertex, store a fixed rest-guide coordinate, nonnegative normalized influence weights and the complete offset in the rest material frame. Retain the longitudinal offset as well as width/thickness. At evaluation, move that offset by the accepted guide center and frame, blend the influences, then apply the placement transform once.

The proposed mapping can be written as:

- `d_ik = transpose(R0_k(s_ik)) * (p0_i - C0_k(s_ik))`
- `q_i = sum_k w_ik * (C_k(s_ik) + R_k(s_ik) * d_ik)`
- `p_i = a_i * p0_i + (1 - a_i) * q_i`

Here p0 is the rest position, C/R are guide centers/frames, s is a fixed rest coordinate, w is an influence weight and a is a mesh-patch hold mask. Guide-root orientation is an upstream constraint. A hard patch hold is not a tunable angular spring, and a soft hold blend can shrink geometry.

Use the same sampler at capture and evaluation; do not reproject vertices to newly nearest guide points after bending. R11's example separates shortest-path director interpolation from unwrapped twist so complete turns are retained. It consumes authored frames: it does not build or validate a physical rod frame, guarantee alignment with the interpolated centerline tangent, or solve contact.

Rest-to-rest reconstruction is a preparation gate. Single-guide, equal-coordinate cross-sections preserve width only under the stated proper-frame and zero-hold conditions. Multiple influences can collapse width; R11 includes an intentional counterexample. Representative stress poses must detect these failures. Imported Skin/cage behavior must pass conversion/pose comparisons before this mapping can replace it. This remains a reference candidate, not a selected universal production evaluator.

For a CPU rod comparator, R13 adds a separate solver-index contract. Rod-property calculation can reverse endpoints and optimization can reorder constraints. Keep semantic guide/segment identities mapped through those changes. Engine-generated transported frames do not recover authored blade width or twist. Build fresh rest settings, derive the ordinary properties, then apply any validated material-frame overrides and recompute consistent relative rest rotations. Prefilling a frame alone can skip derived work; inspect the exact implementation and prove rest reconstruction. Rod data and remaps.

Appearance and geometry parity

Verify that decoding preserves vertex/face association, UV seams, mapping channels, material IDs and intentionally edited foliage normals. Recomputing geometric normals can change the appearance of an asset with authored normals. Rotating every source normal by a guide frame is also not universally equivalent to the normal of a blended deformation. Choose and test the policy with the actual materials and lighting.

R11 distinguishes geometric-normal reconstruction from an explicitly artistic transport of authored corner normals. A weighted rotation blend is not the exact inverse-transpose deformation Jacobian. Preserve corner splits, diagnose near-zero blended normals, rebuild tangents under a declared convention and qualify the result through the actual renderer and Bake path. Its supplied tests cover geometric placement-normal arithmetic, not general authored-normal preservation.

Nonuniform scale, mirroring and shear need an explicit rule. A circular local envelope may become elliptical in world space, and normals require the appropriate transform. The first prototype may reject unsupported transforms with a clear diagnosis; it must not silently assume uniform positive scale.

Invalidation and asset audit

Rebuild preparation when source topology, reference pose, root metadata, membership, guide construction or binding semantics change. UV/material edits can remain appearance-only only if those channels were not used to discover roots, membership, stiffness or other geometry behavior.

A surviving placement ID is insufficient for fit reuse. Its full transform, source revision, preparation revision, relevant obstacles, fit settings and evaluation-time policy must still be compatible. Density, quota, Relax and acceptance changes may alter surviving placement inputs under the current foundation.

Use the asset audit template on a representative set: clean ribbons, disconnected blades, joined-base clumps, precurved blades, opacity cards, broad leaves, scans and proxy sources. Record manual preparation time and rejection causes. Do not turn an untested “usually automatic” claim into an asset support promise.

04 Collision and solvers

Status: inspected dependency behavior, supplied diagnostics, engineering hypotheses and a comparison plan. The R12 source review, R13 review and R14 review distinguish code checks, reported native runs and analytic fixtures. No production solver has been selected or independently executed here.

Neighborhoods and geometry roles

Keep render source, driver geometry, receiver and obstacle proxy distinct. A receiver anchors a plant; it is not automatically an obstacle. Rooted mode uses one-way interaction initially: the fixed stone affects the grass, and the stone receives no reaction motion. Plant-to-plant contact is a separate opt-in scope.

Build a bounded spatial broad phase over accepted plant reach and obstacle geometry. Query potential contacts before narrow-phase tests. Use the union of old/new obstacle influence and previous dependencies during updates. An obstacle moving away matters as much as one moving into contact. A single nearest object or nearest surface sample cannot establish the complete constraint set.

Contact fidelity

The collision representation must match the intended behavior. A convex hull can remove concavities; a terrain hull can bridge a valley; a conservative capsule can block a ribbon that could physically turn edge-on through a gap. Show proxies and their effective margins.

For a thin blade, test whole segments/capsules or an explicitly represented ribbon. Endpoint tests can miss a segment crossing a stone. Finite surface samples can miss crossings between samples. A component fully inside a closed obstacle can require inside/outside classification even when no triangle pair intersects.

After decoding, validate the visible mesh against every relevant obstacle. Guide clearance does not prove mesh-edge or triangle clearance. Smoothing, shell thickness, interpolation and rendering subdivision can introduce new contact. The validation envelope must match the geometry actually delivered.

For signed distance fields, record voxel size, band coverage, sign validity and approximation error. Open meshes, thin surfaces and intersecting obstacles need a declared policy. A narrow polygon gap can disappear in a coarse field. Surface normals alone do not supply robust inside/outside semantics for arbitrary meshes.

Clearance accounting

Define terms once: visible half-width or proxy radius, requested clearance, obstacle padding, discretization error and numerical acceptance tolerance. Avoid counting the same width twice. For a straight parallel channel, if effective radius already includes visible half-width plus clearance, a necessary centerline-channel width is approximately twice that radius; it is not that value plus the blade width again.

That lower bound is not a general feasibility proof. Curvature, base attachment, admissible strain and full three-dimensional routes matter. Opposing contacts must remain separate constraints; averaging their normals can cancel the information needed to escape penetration.

Rest state, initial state and loading history

State	Meaning	Why it matters
Rest shape	Authored preferred shape and material target	Must remain independent of a temporary collision-free initialization
Initial geometry	Starting iterate or simulation pose	May need contact recovery or explicit preparation
Completed fit	Accepted geometry for the declared inputs	Does not automatically redefine the rest shape
Simulation history	Motion, velocities and solver state over time	Can change the settled result and restart requirements

The original use case includes an already intersecting source beside an already fixed stone. Keep that test. Moving the stone in gradually from a clear position is a useful comparison but solves a different loading history.

A deterministic internal continuation or relaxation scheme may be a legitimate fitting algorithm. Record its start and continuation policy without animating the artist's scene or implying a unique physical equilibrium. With a separate straight-rest target, a clear bent initial guide has a restoring load. Rebuilding rest constraints from that bent guide can erase the load entirely.

Numerical candidates

Candidate	Useful first comparison	Main unresolved issue
Geometric contact projection plus shape/length regularization	Small static fit with explicit limits	Deep overlap recovery, multi-contact convergence and shape quality
Rod or position/constraint relaxation	Rooted bend and optional material frame	Discretization, stiffness/work dependence, collision fidelity and initialization
Jolt CPU soft-body/rod route	Mechanics/control comparator using an existing dependency	Checked bare-rod environmental contact samples vertices, leaving segment and root-policy gaps
Ribbon/surface solver	Frame-aware contact for wide or twisting blades	Greater state/contact cost and source preparation
Full rigid body	Pebbles and rigid leaves	Cannot provide vegetation bending or leaf folding

Jolt's CPU Cosserat-rod addition and its later experimental GPU hair path are distinct capabilities. A GPU-path restriction is not a general rejection of CPU rods, and a CPU rod feature is not proof of full continuous capsule collision suitable for this product. Inspect the pinned implementation and run the same contact fixtures before choosing it. Jolt 5.4.0, Jolt 5.6.0.

What R12 resolves at the Jolt pin

The local Lab build pins commit `1b2e77e3c6c09709e5edc79dc208459b8ff32051`; its Core header says 5.6.1. Eleven decision-critical local files match the public files at that commit after line-ending normalization. This is a source check, not a built/loaded-module identity check or an audit of the complete dependency. The exact commit and build flags matter more than a release nickname.

R13's wider source ledger matches 48 local file hashes, while the pinned CMake project string remains 5.6.0. Preserve both observations rather than substituting a nearby tag. Reported build flags and binary/loaded identity still need their own execution evidence.

In the checked bare CPU rod environmental path, shapes receive vertex positions and inverse masses and keep one selected collision plane per vertex. That interface supplies no rod segment or varying blade-width envelope. A segment can cross a stone while both sampled endpoints remain outside. Adding samples can reduce this error but does not prove continuous whole-segment contact. This finding is limited to that path: the engine also has soft-body face handling in its rigid-body CCD path, so it must not be generalized to every Jolt soft-body subsystem. Vertex iterator, rigid-body CCD implementation.

Root construction also needs a controlled test. The normal rod-property calculation uses the minimum endpoint inverse mass for a rod, and its angular corrections use that rod inverse mass. A zero-inverse-mass root can therefore suppress adjacent rod rotation corrections and behave more like a clamp than an intended hinge. Contact projection also skips fixed vertices; it cannot repair a protected root already inside a stone. Compare position-only, soft angular and hard angular attachment explicitly rather than inferring them from one mass flag. Rod properties, rod constraints and projection.

Updated method recommendation

R13 supplies a reproducible diagnostic seed rather than a completed contact adapter. Its reported runs use Linux/GCC, double world positions, a single-threaded job system and no compute backends. Archive integrity, stored results and local source matches were checked here; the engine executions were not repeated.

Reported diagnostic	Recorded observation	Consequence and limit
Segment crossing and stock rod	Endpoints clear; capsule hit; ten rod updates leave the crossing with about 0.000000674 m endpoint drift	Confirms the sampled-contact gap in that setup; a hard-clamped single segment is not a feasible routing benchmark
Closed cube and interior points	Contained capsule has no surface hits; center occupancy false, off-axis interior true; deep soft vertex lacks a plane	Surface contact is not containment; shared-edge ray degeneracy needs an explicit ambiguity policy
Visible width and two walls	Center capsule clear, offset edge hits; diameter 0.10 capsule contacts both walls at gap 0.08 and neither at 0.12	Coverage and local gap detection only; no full ribbon fit or two-stone correction executed
Equivalent tiny crossings	One hit at scale 1; zero at 0.001 and 0.00001, including proportionally reduced collision tolerance	Qualify numerical scale/normalization against an independent analytic control; the rejecting branch was not diagnosed
Far-origin query and arithmetic	Hits at origins 0 and 1,000,000 with matching base offset; float arithmetic loses a 0.001 delta	Subtract a nearby origin before reducing precision; one successful query does not qualify all host transforms
Rigid hull settling	Sleeps after 66 updates with zero error bits; final query penetration about 4.289 mm under a 1 mm slop setting	Sleep and slop are not final visible-clearance certificates; this residual is one query/build observation

The exact stored JSON retains every value. None of these rows is a production performance result or an independent repetition by this library.

R14 uses another recorded configuration: single world precision, SSE2 baseline, assertions, GCC 13.3 and CMake 4.1 with single-threaded jobs and compute disabled. Keep it separate from R13's double-world-position/AVX profile. Its native JSON contains seven records across five families:

R14 records	Reported observation	Consequence and limit
H01, H02, H02b crossing	Capsule hit with clear endpoints; three-point rod still crosses after 60 updates at both 5 and 50 soft iterations; no contact observations	Reinforces the missing witness. The first span is clamped; these are detection controls, not feasible-routing comparisons
H03 containment	Contained capsule hits solid Box but not closed Mesh under either backface mode; cube-center parity sees two equal-fraction hits, off-axis one	Backfaces do not create remote surface contact. Raw no-hit does not itself identify an ambiguous solid query
H04 inside mesh	Free rod stays deeply inside with no contact observations	Requires separate solid classification/recovery or an explicit invalid-start result
H04b inside sphere	Three contacting vertices initially; free rod leaves the sphere and reaches about 11.273 m/s after 60 updates	Dynamic ejection is not a rooted static-fit success or evidence of acceptable settling

R14 records	Reported observation	Consequence and limit
H05 protected root	Root remains near x=0.2 inside the unit sphere while update error bits stay zero	Feasibility and geometry acceptance remain application responsibilities

R14's separate analytic results contain nine fixtures and fifteen reported Boolean checks. Its tiny-asset, large-origin, gap, corner and width expectations are mathematical controls, not additional native solver passes. R13's native tiny-query and far-origin results retain their original scope. Neither study measures throughput or supplies a completed contact fitter.

Track or method	Proposed role	Decision still needed
Jolt rigid bodies	Separate rigid-stone/debris settling track; fixed mesh/heightfield obstacles and suitable convex/compound moving proxies	Residual overlap, admitted capacity, lifecycle and transform publication
Focused static fitter	Leading hypothesis for prepared rooted vegetation; preserve rest/root constraints and test whole segments/envelopes	Recovery, multi-contact, visible quality and complete update cost
Jolt shape queries	Candidate contact backend for that fitter, compared with a direct triangle BVH	Contact completeness, mesh/solid policy, cooking, memory and warm-query cost
Stock CPU rods	Small mechanics/control comparator; document or address contact/root limitations	Whether adaptation is cheaper and more reliable than the focused fitter
Experimental GPU hair	Separate future investigation only if measured needs justify it	Integration, representation, contacts and output support; no current adoption

Query reuse does not supply the fitter, root policy, rest constraints, convergence checks or renderer output. A custom collision listener alone cannot add rod-segment geometry absent from the checked iterator. Compare both contact backends using identical fixtures and acceptance criteria. Shape-query interface.

R14 makes the hook distinction precise. The pinned engine has a public advanced rigid-body contact-generation callback, but rigid pair processing diverts soft bodies into their separate pipeline before invoking it. The soft-body listener adjusts pair settings and observes a const post-solve particle manifold; it exposes no endpoint/frame constraint-row insertion API. Budget a maintained internal extension if a CPU rod route requires that missing seam, or let a focused fitter own response through public queries. This finding does not mean every custom Jolt contact requires a fork. Rigid callback, soft-body listener.

Query and update adapter details

Construct a segment capsule from midpoint, local Y-axis orientation, half the endpoint distance and a declared radius. Arbitrary axial stretching is not valid capsule scaling at this pin. Reject zero-length segments or use an explicit sphere fallback. Keep body-origin and center-of-mass transforms distinct. When the query is shape 1, the documented penetration axis moves shape 2 out, so a corresponding translation of the query uses the opposite normalized direction. A separation-shell query can return negative depth. Prove these conventions with a known primitive before distributing corrections to deformable endpoints. Capsule, query results.

A rod material frame uses local Z along the span; a query capsule uses local Y. Convert deliberately rather than reusing the same quaternion. A raw contact also supplies no semantic material fraction along the guide. Derive and validate that fraction from the capsule-side witness/axis or an independently qualified closest-feature calculation. An arbitrary obstacle-side projection during deep overlap is insufficient.

R14 proposes a local frozen-witness response. For a qualified fraction u, outward unit normal n, obstacle witness c, radius r and clearance h, let p=(1-u)x0+u*x1 and g=dot(n,p-c)-r-h. Its endpoint gradients are (1-u)n and u*n. With mobilities w0/w1, a hard projection uses denominator D=w0*(1-u)^2+w1*u^2 and multiplier -g/D when g is negative. Preserve a pinned endpoint with zero mobility; negative gap with D=0 is an immovable conflict. This is a local linearization to test, not a complete solver. Width offsets add frame/rotation gradients; every correction must refresh witnesses and revalidate roots, rest/length and all contacts.

Whole-segment coverage is spatial coverage at a pose. A translation cast of a rigid capsule does not cover arbitrary endpoint bending, rotation or ribbon deformation through time. Declare final-pose fitting separately from continuous motion/route preservation. Both need an independent final visible-shape validator.

R13 also clarifies contact refresh. In the checked CPU soft-body path, collision planes are selected before integration and reused across its soft substeps. Increasing soft-body iterations changes integration/constraint work without automatically refreshing those planes or adding segment geometry. World collision steps and manual CustomUpdate calls are different boundaries; avoid double stepping a body through both paths. Root inverse mass zero still needs canonical zero initial velocity to be stationary. Free/finite angular attachment remains a prototype, not a reported success. Runtime/update contract.

Physical scale and solid/surface policy

Declare conversion from scene units to the engine's physical scale and choose a local numerical origin. Test equivalent scenes at different scene-unit settings and translated far from the world origin. Engine tolerances and default vertex radii are numerical defaults, not calibrated grass clearance presets. Record all effective values; do not silently retune global engine settings per blade.

The checked triangle helper has finite backward thickness. A closest triangle or its backface slab does not classify every point deeply inside an arbitrary closed solid. Keep a separate declared containment policy and distinguish closed solids from open/sided surfaces. A root-conflict preflight should preserve the plant and report the conflict. Triangle helper.

Do not replace that policy with unqualified raw mesh parity. The checked fallback counts triangle hits along a positive-Y ray; the report's cube-center ray meets a shared face diagonal and returns an inconsistent inside result. Use independently checked solid classification with edge/vertex handling and an ambiguous outcome. A no-hit surface query cannot certify that an entire blade is outside. Point classification implementation.

For static geometry queries, normalization into a qualified local metric is a candidate response to tiny-scale failures; it must pass mapped-back acceptance tests. Dynamic rescaling also affects gravity, velocities, mass/inertia, compliance and time. Numerical convex rounding, artist clearance, length-valued collision tolerance and dimensionless penetration tolerance remain different quantities. Reducing one tolerance is not a complete scale policy.

Shape names are not body-role rules. An immutable StaticCompound can form a moving body when its children are eligible. Triangle meshes have a documented dynamic/kinematic exception with supplied mass properties, but mesh-mesh and mesh-heightfield contact remain unsupported; use convex/compound moving debris as the first rigid profile. These engine possibilities do not extend Cyrus support automatically. Mesh roles.

An SDF backend would add sign, sampling, error, cache and out-of-band policies. Under a true 1-Lipschitz distance function with bounded field error, samples including both endpoints with maximum straight-segment spacing h support a lower bound of minimum sampled distance minus h/2 minus field error. An arbitrary interpolated or narrow-band field does not inherit that guarantee. This is an analytical condition, not a supplied field implementation or benchmark.

Convergence, failure and budgets

Track root error, frame error, length/strain change, visible-mesh penetration, residual motion if applicable and work consumed. “Reached frame 90” and “object sleeping” are insufficient acceptance criteria. Calibration must retain topology, mass mode and units so a refinement does not accidentally change the experiment.

Bound affected placements, guide samples, obstacle geometry, candidate pairs, contact storage, iterations, scratch and output geometry separately. Report limit exhaustion distinctly from setup failure or unresolved contact. Neither raising iterations nor reducing friction repairs missing collision geometry or contradictory constraints.

Use the same units, input geometry, decoder and acceptance metrics across comparator methods. Solver-specific coefficients need not have the same physical meaning. Performance comparisons must include preparation, neighborhood search, solve, decoding, output and publication rather than reporting only the integration loop.

Check engine update error bits as well as completion. At this pin, body-pair/contact/manifold capacity errors can mean contacts were ignored. An ordinary return from an update cannot by itself admit the result for publication. Record the exact error, keep the previous complete result and distinguish capacity failure from unresolved geometric contact. Update errors.

The eight solver decision experiments translate R12 into bounded future comparisons. Every experiment remains Pending.

Experimental compute hair can run without scalp data in an official source example; scalp authoring is optional. This does not remove its separate compute/material/output integration or its sampled environment-contact limitations. No compute-hair runtime was executed by R14 or this library. No-scalp collision example.

05 Architecture and output

Status: current-source observations followed by proposed integration design. The snapshot is a dirty working tree on 24 October 2026, not a clean release. File hashes and branch/HEAD are in repository-snapshot.json.

Current source boundary

Observation	Inspected source	Applicability
Native placement Record is 96 bytes and contains transform, source, receiver and identity data	placement_store.h	No per-placement vegetation deformation payload exists in this record
Retained Mesh presentation uses reusable source geometry and per-instance presentation data	mesh_display.inc	A new shape evaluator needs an explicit output path
Standalone Physics Lab creates Jolt rigid bodies and static collider bodies	jolt_world.cpp	Existing rigid workflows do not establish a rooted-rod implementation

Observation	Inspected source	Applicability
Working-tree Lab documentation is 0.3.0; the committed historical baseline differs	Lab README	Preserve the distinction when interpreting older reports
Jolt is pinned by exact commit in the Lab build tool	build.py	A version feature description does not establish local integration of that feature

The current README reports Scatter 0.86.9.2/schema 59. R03's public-main version assumptions do not replace this local baseline. No host or renderer qualification was performed for this documentation task.

R12 inspected upstream Jolt but could not audit the local Lab wrapper from the public repository. Our narrower local checks confirm the dependency pin, rigid-body construction and declared limits; they do not qualify wrapper cache ownership, cancellation, Undo or Scatter integration. Keep the dependency findings separate from the wrapper's acceptance status.

Proposed pipeline

1. Finish ordinary generation, projection, Edit handling, spacing/Relax and acceptance using existing ownership rules.

2. Capture a compatible accepted placement generation and owned source/obstacle snapshots on the host boundary.

3. Identify affected placement IDs and fit their prepared drivers using copied numeric data.

4. Decode and validate visible shapes under declared work/memory limits.

5. Stage a complete interaction result bound to the exact placement generation and dependency revisions.

6. Publish atomically, then let viewport, renderer and Bake consume that completed result.

No later ordinary placement cleanup may silently move a planted root after fitting. Any upstream change that affects roots or source transforms invalidates the downstream fit. Rigid settlement needs a separate publication inlet because its completed transforms intentionally differ from ordinary projected placements.

Reuse boundary

Existing capability	Candidate reuse	Feature-specific responsibility
Owned input snapshots, units and stable-ID conventions	Common numeric capture and identity rules after source review	Root/rest metadata, semantic binding and full dependency revisions
Cooked rigid shapes and engine queries	Obstacle acceleration or capsule/shape contact queries	Whole-segment/envelope selection, multi-contact and containment policy
Worker execution	Bounded native work using copied inputs	Job/barrier audit, cancellation boundaries and independent memory limits
Completed rigid transforms	Rigid-settle results keyed by Cyrus identities	Rooted AcceptedFit state, quality checks and mode-specific publication

Existing capability	Candidate reuse	Feature-specific responsibility
Existing mesh/render/Bake infrastructure	Reviewed host adapters and retirement conventions	Per-placement shape evaluation, appearance channels and actual output parity

These are reuse candidates, not a claim that the current Lab implementation already supplies every listed contract. The proposed PreparedSource, AcceptedFit and EvaluatedVariant roles remain engine-independent. Jolt body IDs and temporary contact-array offsets cannot replace stable scene identity.

Ownership and data

Keep shared source preparation immutable. Store optional fit state for affected placements keyed by owner, stable placement identity, source/preparation revision and compatible placement generation. A compact sparse deformation table is a proposed starting point; it is not a mandate to enlarge the existing 96-byte placement record.

The result should be engine-independent: rest/source references, accepted driver/frame state, status/diagnostics, deformed bounds and the revisions needed to validate use. Do not persist solver pointers, body handles or worker-array offsets as scene identity. An optional resumable simulation state needs a different, versioned contract from a static accepted shape.

R11 usefully separates PreparedSource, AcceptedFit and EvaluatedVariant as proposed record roles: immutable rest/binding data, placement-specific accepted guide state, and evaluated output geometry/bounds. These names describe ownership boundaries, not delivered C++ types or an approved persistence schema. Bone/cage authoring tools may feed the chosen representation only after conversion and pose parity are demonstrated.

Place bulk storage, geometry math, contact search and numerical work in native C++. Keep UI and bounded authored commands in the existing orchestration layer. Host API access, source evaluation and publication follow the host-thread boundary; numerical workers use owned snapshots. A renderer callback may have its own execution/lifetime constraints and must consume pre-evaluated owned data safely rather than assuming it can access arbitrary host state.

Invalidation and reuse

Change	Proposed response
Obstacle transform/geometry or membership	Reconsider old/new influence and previous dependents; refit compatible affected placements
Accepted placement/source transform	Refit surviving IDs whose relevant inputs changed
Source topology, root metadata or binding	Rebuild preparation and invalidate dependent fits
Appearance-only edit	Reuse shape only when appearance channels do not affect preparation/physics
Fit quality, clearance or base attachment	Refit the affected interaction set
Selection, camera navigation, unrelated animation	Reuse completed static result and unchanged buffers

Change	Proposed response
Relevant animated input in Live mode	Re-evaluate according to existing Live semantics and the declared time policy

Before reuse, compare full compatible inputs. Stable IDs alone do not guarantee unchanged accepted populations, transforms or source assignments under F02. A local edit can affect an entire guide, and plant-to-plant coupling would broaden the dependency set.

A cooked-shape key must include geometry revision, cooking/representation policy and the supported scale policy. Translation/rotation can often reuse the same local shape while updating its pose; geometry changes or scale baked into vertices require a new compatible shape. This is a cache-design requirement to verify, not a claim that all current wrapper caches obey it.

R13 further distinguishes shape cooking from transformed query instances. Include role, units, winding, convex rounding and cooking tolerances in compatibility. Engine ShapeSettings cache Create results; changing settings without clearing/replacing that cached result can reuse old geometry. Prefer immutable per-generation snapshots and a newly compatible cooked result rather than mutating a shape while another job queries it. These are integration requirements, not evidence of a fault in the current Lab cache.

R14 closes a further ownership gap: per-query locking does not freeze the obstacle set for an entire iterative solve. Pin one immutable query-world generation, private world or owned transformed-shape snapshot for all iterations. A stone edit creates another generation. Rejecting stale publication is necessary but cannot repair a solve that mixed old and new obstacle poses internally. Refcounting preserves lifetime; shape and transform revisions preserve meaning.

Within a shared Jolt runtime, multiple worlds can share compatible immutable shapes; bodies, IDs and constraints remain world-specific. Factory, dispatch/type registration, default material, allocation and diagnostic hooks have shared lifetime ownership. A new fitter must not replace or unregister those services while the rigid service uses them. Inspect the actual Cyrus module/linkage, ABI/build and allocator boundary before sharing engine resources; matching a source SHA alone does not establish that compatibility. This is a required integration audit, not qualification of the current wrapper. Multiple-system contract.

Rod preparation, reset and result extraction

A CPU rod experiment needs canonical positions, velocities, rod orientations/angular velocities and immutable rest constraints. Resetting only particles, or hard-skinning them, does not reset the complete rod state. At this pin, ApplySoftBodyCreationSettings requires removal from the physics system before reinitialization; canonical recreation is another baseline. Treat engine replay as a distinct state/membership contract, not as an arbitrary rest-frame editing API. Body reset contract.

Preserve semantic guide/segment remaps through preparation and optimization. Extract completed particle positions and rod rotations in the body's current center-of-mass frame, then apply that frame exactly once. Rigid body-origin and COM transforms differ too; test an offset pivot before publishing settled placement transforms. The source, obstacle, basis/origin/scale and binding revisions belong to the immutable result signature. COM and transform conventions.

Choose world Update or the documented manual CustomUpdate route deliberately, with one owner of stepping and supported locking/lifetime boundaries. A post-step particle projection must also own contact refresh, bounds and velocity/frame consistency; mutable vertex access alone does not qualify it as a safe contact adapter.

One visible shape across outputs

Build one deterministic CPU reference evaluator for the selected representation, then use its completed geometry for the first viewport, render and Bake experiment. This is an architectural recommendation, not a requirement to eagerly expand every unaffected instance. Prove one explicitly selected production renderer first. No renderer has been selected or qualified by this research packet; record its ClassID, exact build, CPU/GPU mode and adapter version when that experiment is authorized.

Max's GeomObject/render interfaces and render-time instancing facilities provide relevant hooks. They do not automatically give each Scatter placement a distinct deformation through a shared source callback. Mesh ownership, validity intervals, temporary mesh deletion and concurrent reader lifetimes need an explicit adapter design. GeomObject, render-time instancing.

Validate UVs, map channels, material IDs, smoothing/explicit normals, bounds, picking and frozen output. A correct viewport silhouette does not qualify renderer output. Render-time subdivision or displacement requires a stated support boundary.

R11 adds important adapter checks. Max Mesh shallow copies can share channels, and a renderer can attach mutable mesh cache data. Owned numerical snapshots and consumer-specific SDK adapters therefore need separate lifetimes; wrapping a borrowed Mesh in a smart pointer does not make its channels immutable. Mesh ownership/cache API.

Capture a coherent evaluated mesh and transform before traversal, including object offsets, transformed parents and world-space modifiers, to avoid double transformation. INode evaluation and transforms.

The explicit-normal SDK documentation contains a limited-pipeline-support warning. Treat that as a reason to test the selected custom-object, ordinary-mesh and Bake routes; it does not establish that every current renderer ignores authored normals. Use a visibly non-geometric normal fixture. When baking a reflected transform, permute winding and matching map/normal corner indices together and preserve tangent handedness. MeshNormalSpec.

Picking must use the completed bent geometry and a batch-face to placement/source-face map. Test a bent tip outside its original bounds. Full mesh delivery, exact repeated variants and renderer-proxy export are separate qualification profiles. A proxy conversion command can help prepare an owned source; it is not a generic runtime extraction or arbitrary deformation API.

Memory and publication

Let A be affected placements and V the render vertices per source. Unique expanded positions alone can scale with A times V, even if the guide state is compact. This is a budgeting observation, not a measured allocation or performance claim. Include shared rest data, preparation, guide state, contact scratch, decoded geometry, old/new publication overlap, renderer copies, GPU buffers and Undo retention in peak accounting.

Exact identical shapes can share a deformed prototype. Approximate shape clustering requires an error budget and contact revalidation at each placement; otherwise a reused variant can intersect a stone. Track variant count and temporal/LOD popping before treating clustering as a solution.

An exact variant key should include source revision, evaluator version, effective local guide pose, normal policy and geometry-affecting options. Confirm equality after a hash match. Batching reduces submission overhead while distinct position/normal arrays still require storage; count geometric positions, split output-stream vertices and faces separately.

Cancellation, stale inputs, allocation failure and owner deletion must retire staged work without late publication. Keep previous complete output available. Readers need generation leases or another proven lifetime mechanism

while old results retire. Save/load and Undo should restore coherent authored settings and results without holding an ordinary Undo transaction open across asynchronous work.

For engine reuse, make each update bounded and check cancellation between calls; do not assume a safe abort midway through a running Jolt update. Audit nested scheduling and barrier waits before sharing a host worker pool, including oversubscription and shutdown. Preserve separate outcomes for setup failure, cancellation, engine-capacity errors, unresolved contact and accepted quality. These are proposed adapter obligations, not measured cancellation guarantees.

06 Houdini and tyFlow workflow comparison

Status: documented building blocks and unexecuted recipe proposals. R09 HTML and R10 PDF are the latest supplied practical study. Use their detailed node/operator order for reproduction, with the limitations below. R05-R07 remain preserved earlier research.

How to read the vendor evidence

The latest study separates documented controls (D), saved example structures reportedly inspected by its author (I), assembled recipe proposals (P), and executed DCC tests (X). It reports no X-level Houdini, Max/tyFlow, renderer or round-trip tests. Downloaded-asset hashes and arithmetic fixture checks are useful provenance; they do not establish simulation success. This consolidation has not independently repeated the author's HDA decompression.

The HTML has 14 sections. The PDF is a 64-page presentation of the same study, with cover/contents, page furniture and some presentation/text differences. Both originals are retained; they are not treated as identical files or independent corroborating experiments.

Corrected comparison

Scenario	Houdini candidate	tyFlow candidate	Qualification limit
One fixed stone, rooted grass	H1B static Guide Collide; H1C separate rest/current Vellum setup	C clear guide with Particle Bind/Angle Bind and fixed stone	Full roots, frames and decoded mesh need checking
Two fixed stones	H2B joint VDB candidate; H2C joint fixed-collider Vellum	C2 joint guide bindings and both colliders	Simultaneous contacts; no guaranteed unique route
Conditioned grass	H1A/H2A moving collider then hold	A/B direct ribbon with moving collider then hold	Explicitly history-dependent loading
Rigid pebbles	Packed geometry with RBD Configure/Bullet	PhysX Shape plus PhysX Collision	Clear initial bodies, correct hulls and terrain
Rigid leaves	Thin rigid proxies and transformed render source	TF-LR rigid PhysX route	Shape remains rigid; cannot fold
Deforming leaves	H4D Vellum cloth surface	TF-LD Cloth Bind with CCCS	Unique deformation, thickness and cache/output validation

See the corrected planning matrix. It is a synthesis, not a replacement claim that the recipes were run.

Three distinct evaluation models

A static geometry operation recooks from its current input without a normal timeline simulation. A frame-local internally settled result still performs integration/relaxation inside the evaluation. A timeline simulation uses earlier states and can later be frozen or cached.

Houdini's Vellum Simulation Type Quasistatic rolls current inputs forward internally. H22's Orientation Solve Quasistatic is a separate hair bend/twist control. Guide Deform Preserve Shape is another distinct relaxation feature; its documentation does not establish obstacle contact. Avoid merging the three meanings into a claim of an automatic universal fitter. Vellum Solver, H22 Vellum changes, Guide Deform.

tyFlow's history-independent mode restarts at each frame; that description does not establish equilibrium fitting. A held cached frame can be a static deliverable from a history-dependent simulation. Main settings, reset behavior.

Root and decoder details that matter

The latest Houdini fixed-stone recipe explicitly keeps straight-rest constraints separate from a clear bent initial guide and reinitializes its orientation. This is a substantial improvement over assuming that an already bent rest guide will spontaneously seek the stone at zero gravity. It remains an assembled, unexecuted scene proposal.

Its named Path Deform route is a practical decoder candidate for straight prepared strips. A constant-up geometric frame does not transfer arbitrary simulated material twist and can fail near a parallel tangent. The two-stone recipe therefore retains a guide-level qualification boundary until the visible strip's frame and contact pass. Path Deform.

tyFlow's guide route uses Particle Bind and Angle Bind. Hair Bind attaches particles to Ornatrix strands; its name is not evidence of a general grass solver. Bind-segment collision must be enabled explicitly, and the Collision operator's absolute radius supplies its object envelope. A root particle's rotation is not equivalent to a solved rod material frame. Hair Bind, Particle Bind, Rotation.

For detailed meshes, isolate blade association and inspect every decoder. A good low-resolution driver is not proof that tyParticleSkin, spline meshing or another binding preserves final contact.

Versions and prerequisites

As checked on 24 October 2026, current SideFX pages identify Houdini 22 and tyFlow's release history lists 2.101 dated 8 October 2026. Earlier minimum node versions do not imply every current parameter existed then. Keep exact build and node versions in each experiment. SideFX documentation, tyFlow release history.

Current tyFlow download/release information specifies CUDA 12.8, while older system/FAQ pages contain older requirements. The general bind solver and CCCS have different CPU/GPU boundaries. FREE license restrictions also affect GPU acceleration and tyCache export. Verify the selected package, driver and license rather than copying an older requirement into a support promise. Download, system requirements, FREE license, CCCS FAQ.

Local project development and host validation remain Max 2027 only. A vendor's broader support does not authorize a different host for our tests.

Reproduction checklist

Record physical geometry, coordinate conversion, scene units/fps, original rest state, clear initialization or conditioning history, root/frame controls, solver budgets and inherited defaults. The latest report deliberately avoids assigning a guessed SI meaning to tyFlow's native gravity values; use its documented real-world-value action and record the resulting value.

Run one blade/body first. Show the driver or hull beside the visible mesh. Then run two obstacles together, move an obstacle away, reset from the declared state, freeze/cache, reopen, render and export/reimport. Keep reset, in-memory hold, disk cache and baked mesh distinct. Cache loading can intentionally continue showing old output after upstream edits.

The written task counts are planned task groups, with unequal preparation boundaries between recipes. They are not measured clicks, artist time, performance rankings or product UX targets.

Missing companion data

R09/R10 refer to common_grass_initialization.json, also named CyrusScatter_Common_Grass_Initialization.json, and to an example-download manifest. Neither was supplied among the ten files, and neither JSON filename was found in Downloads during this organization. The study prints some construction directions and arithmetic results, but that does not restore the missing artifact's exact bytes or provenance.

Retain the recipe IDs and report the dependency as missing. A later reconstruction can be useful if explicitly labeled newly derived fixture data. Do not present it as the author's original or a validated .hip/.max scene.

07 Validation and prototype plan

Status: proposed experiments; every implementation/host row starts Pending. This plan is available for a later authorized task. Documentation generation and integrity checks are recorded separately in VERIFICATION.md.

Gate 0: fix the experiment boundary

Select one prepared source, fixed obstacle fixture, source/rest convention, supported transform range and one production renderer/build. Define numeric acceptance tolerances in physical units before running. Record clearance, permitted strain, root position/frame error and output comparison metrics. Candidate values in research remain starting hypotheses until calibrated.

Keep an original-intersection fixture for the user's actual request. Add a collision-free initial guide and a moving-obstacle conditioning variant as separate experiments. No pass in the conditioned branch substitutes for a fixed-from-start overlap-recovery result.

Use owned disposable scenes/profiles and exact matched script/native identities. Max 2027 is the development/qualification host. Preserve artist sessions and Test Scene assets. Offline arithmetic or source inspection must not be reported as a Max pass.

R13 supplies an authored headless harness and reported Linux results. Preserve those failures as regression seeds; rerunning them through a future Cyrus unit/snapshot adapter is new work. For primitives, use analytic geometry as an independent reference. For meshes, separate surface intersection from solid classification and test edge/vertex degeneracy. The same query under evaluation cannot be the sole acceptance oracle.

R14 adds seven reported native records across five families, plus nine analytic fixtures with fifteen stored checks. Its single-world-precision/SSE2/assert build differs from R13's double-world-position/AVX build. Keep separate receipts and distinguish native physics, arithmetic expectations and proposed solver response. R14's E1-E7 are report-local gates; they do not share the meanings/status of our R12-E1 through R12-E8 identifiers.

Gate 1: output feasibility and asset audit

Run R12's smallest source-falsification probes alongside this gate: exact dependency identity, bare-rod segment crossing and root attachment behavior. They can use a tiny native harness without waiting for a large Max integration. They do not replace the authored output-parity experiment below. The eight decision experiments map these questions to the shared validation matrix; none has been executed here.

Create one authored deformation through a minimal CPU reference decoder and carry it through viewport, chosen final renderer and Bake. Compare actual positions/bounds, root registration, UVs, material IDs and explicit normals. This gate can precede a sophisticated solver and can expose an unsuitable output design early.

Audit representative assets at the same time. Record topology families, necessary manual work, root/frame ambiguity, unsupported proxy access and binding errors. Do not build a broad automatic preparation product before measuring this asset burden.

Use R11's embedded public/synthetic examples as small representation fixtures. Distinguish our independently recounted topology from its supplied seven topology and ten evaluator result records. All Max/renderer gates remain pending. Add explicit two-sided faces, independent coincident blades, shared vertices, full-turn twist and opposing-frame blends before generalizing the decoder.

Exit evidence: a matched output fixture plus actual audit records, or an explicit failed gate identifying the representation/output limitation. The current research library contains no such execution evidence.

Gate 2: contact and numerical comparison

Run the same prepared blade/stone fixtures through a focused fitter and a rod/constraint comparator. Compare direct triangle-BVH contact with Jolt shape queries inside the fitter, keeping the same source, decoder, physical scale and acceptance criteria. Include deep initial overlap, segment crossings between samples, visible-width contact and root conflicts. Do not count contact-query time alone as the solver or full-update cost.

R13's clamped one-segment crossing is a detection test. Add a feasible multi-segment detour and implement the required rod contact adapter before comparing total correction quality or speed. Measure canonical reset, root orientation, authored frames/remaps and collision-refresh cadence. Specify whether acceptance covers only the final static pose or a continuous deformation path.

Retain R14's 5/50-iteration three-point crossing as another detection seed, and its free-rod ejection as a failed static-fit control. For the first response experiment, provide root freedom and enough guide length for a demonstrably feasible route. Test local witness/endpoint correction with an independently known parameter, capsule-Y/rod-Z frame conversion and an immovable-contact denominator. Every rejected or ambiguous fixture needs a feasible companion that must be accepted; blanket rejection is not product success.

Then test two fixed obstacles in one solve. Include a narrow channel, a possible out-of-plane route, contradictory protected-root constraints and symmetric alternatives. A failure should identify unresolved contact or work exhaustion; do not infer global infeasibility from a local failure.

For a relaxation method, refine timestep/iterations and discretization separately. Measure residual motion, strain and contact error; reaching a chosen frame is not convergence. Record whether results depend on initialization, obstacle approach order or previous fitted state.

Gate 3: lifecycle and product behavior

Move an obstacle into contact, across the vegetation, away from contact, then delete it. Include old/new influence and released plants. Compare recomputation with a fresh setup using the same declared rest/initial state.

Exercise Manual pending edits, relevant Live edits/animation, unrelated animation, selection/navigation, owner switching, cancel, stale completion, deletion, scene reset, save/reopen and Undo/Redo. Check that failures retain the prior complete publication and that its status honestly reports staleness.

Move an obstacle between repeated queries while a job runs: that job must keep one coherent geometry/pose generation even though its successor becomes stale. Also exercise rigid and fitting services starting/stopping together under the actual shared-runtime/module ownership. A safe individual query or a final revision check does not prove either lifecycle contract.

Test placement-input changes: source swap, density/quota, masks, Relax, transform edits, relevant source topology and appearance-only changes. Reuse should follow compatible full inputs, not stable IDs alone.

Gate 4: bounded scale and memory

Measure cold preparation, warm update and no-op reuse separately. Scale affected and unaffected populations independently; a mostly unaffected field must not pay full deformation/output cost. Increase contact density and source vertex count independently of placement count.

Capture neighborhood candidates, active contacts, guide samples, solve work, decoded vertices, output variants, cache reuse, uploads, wall time and actual responsiveness. Account for retained source data, staged/published overlap, render copies and Undo/Redo history. Test allocation/budget failure and resource retirement.

Add unit/local-origin invariance, engine error-bit handling, job/barrier safety and cancellation-latency measurements before accepting engine reuse. The current Lab declares a 5,000-body default/admitted profile. R12's proposed 10,000-body comparison exceeds that profile and requires a separately scoped harness/admission decision; this research update changes no limits or settings.

A GPU or approximate variant path is justified only if these measurements identify a bottleneck and the proposed optimization preserves acceptance. Report absolute memory and complete-pipeline timings with hardware/build/input identity.

Required fixture families

Family	Key cases
Basic shape	Straight ribbon, precurved blade, two disconnected blades, joined-base clump
Contact	One stone, two stones, sharp corner, thin gap, open mesh, deeply embedded component
Anchoring	Position-only hinge, fixed tangent, full base frame, soft angular attachment, root conflict
Geometry delivery	Driver-clear/mesh-penetrating case, UV seams, authored normals, materials, Bake/render parity
Preparation identity	Render/logical/semantic IDs, opposite-winding pairs, independent coincident blades, shared-base policy
Decoder stress	Rest reconstruction, unwrapped twist, opposing blend collapse, authored rig conversion and mirrored corner channels

Family	Key cases
Lifecycle	Into/out of contact, obstacle deletion, relevant animation, Manual pending, stale/cancelled work
Resources	Many unaffected placements, dense local contact, high-vertex source, failure during publication

The validation matrix expands these into individually trackable rows. The experiment record defines the evidence to attach to each run.

Record renderer profile qualification separately from source preparation readiness. Supplied JSON saying an arithmetic check passed, or recording an external headless run, does not change a Cyrus integration row to Pass. The matrix now contains 68 proposed cases: the prior 64 plus four query-generation, runtime-lifetime, witness-response and ejection controls from R14. All remain Pending. R13/R14's report-local gates refine the existing eight decision experiments rather than creating another execution ledger.

What a completed result must report

State exact source/package/loaded identities, scene and dependency hashes, host/renderer versions, hardware, initialization and changed parameters. Attach measured values, diagnostic IDs, screenshots/renders, output files and command/test logs as relevant. Separate numeric, host automation, physical artist interaction and renderer evidence.

Report pass/fail for the declared criteria, untested rows, unsupported geometry and remaining risks. A future installer is required when delivering a new Scatter version for user testing under the repository workflow; this documentation packet is not that delivery.

08 Engineering guidelines

Status: proposed feature guidelines, subordinate to existing repository contracts. These identifiers are local research review aids, not additions to the foundation's F01-F12 contract IDs. A later implementation should resolve them in its scoped plan and acceptance receipt.

Reviewable requirements

ID	Guideline	Evidence needed later
VG01	Preserve accepted placement identity, count and planted root transforms during rooted fitting	Before/after placement comparison; explicit upstream-change cases
VG02	Keep source/rest geometry immutable and preparation reusable per compatible source revision	Source mutation checks and revision-driven invalidation
VG03	Keep root position, tangent and cross-section orientation distinct	Hinge/clamp/soft-base fixtures with measured errors
VG04	Define explicit obstacle membership independent of placement collision scopes	Scope isolation and ground/receiver tests

ID	Guideline	Evidence needed later
VG05	Include every relevant local obstacle, whole-segment contact and final decoded mesh validation	One/two-obstacle, missed-segment and width fixtures
VG06	Declare rest state, initialization, history and repeatability policy	Fixed-intersection, clear-initial and conditioning comparisons
VG07	Reconsider old/new influence and previous dependencies after obstacle edits	Move-away/deletion tests and fresh-setup comparisons
VG08	Reuse only when full inputs remain compatible	Transform/source/preparation/settings/revision tests
VG09	Keep host API access at the host boundary and workers on owned numeric snapshots	Lifetime, cancellation, owner deletion and stale-result tests
VG10	Stage and publish a complete coherent successor; retain prior complete output on failure	Failure before/after commit, stale completion and reader lifetime checks
VG11	Preserve existing Manual/Live and passive browsing behavior	Pending edits, relevant animation, idle/selection/upload counters
VG12	Use one qualified shape evaluation policy across viewport, render and Bake	Actual output comparison in a selected Max 2027 renderer
VG13	Bound computation, geometry, memory and history independently	Peak/retained allocation records and budget-exhaustion cases
VG14	Expose unresolved contact, stale state and limitations honestly	Selectable diagnostics, partial-acceptance and user recovery tests
VG15	Preserve geometry appearance channels deliberately	UV/material/smoothing/explicit-normal fixtures
VG16	Keep static fitting, rigid settling and deformable drape separate in data and behavior	Mode-specific inputs, time/reset and output acceptance
VG17	Preserve render-corner, logical-position and semantic-part identities independently	UV/normal splits, opposite winding and coincident-independent fixtures
VG18	Require rest reconstruction, frame/twist and shared-base compatibility before accepting preparation	Full-turn twist, opposing blends, junction stress and rig-conversion pose comparisons
VG19	Qualify owned mesh adapters and renderer profiles independently from source readiness	Shared-channel/cache lifetimes, explicit normals, transformed sources and exact-variant inspection
VG20	Reuse solver/query capabilities only within inspected and measured boundaries	Exact source/build identity, whole-segment/multi-contact fixtures and same-input contact-backend comparison
VG21	Declare engine scale, origin, capacity and cancellation behavior before adoption	Unit/translation invariance, error-bit admission, bounded updates, job/barrier and shutdown checks
VG22	Qualify the geometry/query adapter with independent truth and final acceptance	Tiny-scale/parity/sign/COM controls; sleep-residual counterexample; shared false-negative detection
VG23	Preserve complete rod preparation, state and step ownership	Rest frames/remaps, root velocities, complete reset, contact-refresh cadence and world/manual-step isolation

ID	Guideline	Evidence needed later
VG24	Own coherent query generations and shared runtime services explicitly	Mid-solve obstacle edits, compatible shape/pose leases, actual module/allocator boundaries and overlapping-world shutdown checks
VG25	Qualify local contact witnesses and their response model before composing a solver	Capsule/rod axes, semantic fraction, endpoint/rotation gradients, pinned mobility, immovable conflicts and refreshed final validation

Before implementation

Define the owner and lifetime of every new setting, source preparation record, obstacle snapshot, worker job, result, renderer mesh and Undo reference. Identify units/coordinate frames, invalidation triggers, unsupported transforms and persistence. Estimate peak storage including old/new overlap and prove the failure path before raising population limits.

Read the current foundation, workflow and relevant source. Research reports and this library do not substitute for a new source/status check. Use C++ for placement-sized numerical work and storage; keep bounded UI coordination in the established scripted/managed boundary unless measurements justify a different choice.

During experiments

Change one meaningful variable at a time, preserve exact fixtures and distinguish setup errors from solve errors and output errors. Store numeric measurements with their full tolerances and units. A selected attractive image is insufficient to establish reliability. A documented operator can support a proposed recipe while that complete recipe remains untested.

Compare the simplest sufficient method with the chosen comparator. Do not add a generic framework, GPU backend, universal DQ skinning system or compatibility layer merely because future research mentions one. A measured limitation should drive each extension.

At handoff

Explain actual behavior, changed ownership, exact evidence and remaining acceptance. Keep independently versioned products separate. Record new qualification in the appropriate existing implementation/ledger location only when authorized work produces it. Preserve historical reports, failed cases and original source hashes.

For external research, request runnable minimal scenes and measurements for specific unresolved questions. Another broad literature survey is lower priority than a reproducible fixed-obstacle/initial-overlap case, a real asset audit or an output-parity experiment.

Documentation updates

Preserve new originals byte-for-byte and extend the source manifest with new IDs. Reconcile contradictions in Chapter 09 before adopting claims elsewhere. Cite primary sources for vendor facts and local source for Cyrus behavior. Label copied experimental settings as proposed; do not silently promote task counts to timings or stored example records to executed tests.

Edit the numbered Markdown, update the checklists, regenerate HTML/PDF and rerun package integrity/layout checks. A documentation pass does not refresh the knowledge corpus, build the website, qualify the plugin or authorize publication.

09 Evidence, corrections and open questions

Status: consolidated review as of 24 October 2026. This register governs how claims from the supplied reports should be used. It is a research decision aid, not a record of executed vegetation simulations.

Evidence categories

Category	What it establishes	What it does not establish
Local source inspected	Structure/behavior visible in the recorded files	Loaded build identity, host execution or renderer support
Vendor documentation checked	Public feature/control described at the access date	A full assembled scene's success or internal proprietary algorithm
Reported example inspection	The research author's account of saved network records/hashes	Independent repetition here or execution in the current DCC
Proposed recipe/design	A specific hypothesis with a procedure	Proven reliability, speed or asset coverage
Arithmetic estimate	Cost or geometry calculation under stated assumptions	Runtime memory, convergence, application contact or performance
Supplied offline check results	R11's stored seven representation and ten evaluator results, with scripts and inputs	Independent repetition of those scripts here, automatic preparation or Max integration
Supplied headless execution records	R13/R14 harnesses, reported Linux CPU diagnostics and distinct recorded build profiles	Independent repetition here, Windows/Max behavior, full fitting, renderer output or performance
Independent data recount	This update's OBJ/GLB topology counts and archive integrity checks	Botanical identity, a contact result or renderer qualification
Cited-source hash checks	R12's eleven local/public files and later supplied ledgers checked against local source after line-ending normalization	The entire dependency, local wrapper lifecycle, binary identity or runtime quality
Executed application result	Only an actual run with inputs, build and evidence	Untested variants, later builds or a whole release

Multiple reports may draw on the same documentation and each other's prompts. Agreement is useful synthesis, not an independent experimental replication.

Source weighting

R01/R02 supply useful engineering and product framing. R04 extends contact, evidence hygiene and validation coverage. R08 is the earlier source-preparation study. R11 supplies a more precise preparation/output contract, two public assets, synthetic counterexamples and a candidate framed-offset evaluator. R12 adds exact-pin solver/contact inspection and eight decision experiments. R13/R14 extend that work with different reported headless probes and more precise query, response and lifetime requirements. R14 is the final supplied study; it supplements rather than replaces R13's distinct results. R09/R10 remain the preferred vendor workflow reading set because they address earlier initialization, framing and version problems. R03 and R05-R07 remain useful historical inputs but must be read with the corrections below.

No report author demonstrated a complete Cyrus rooted-contact implementation. R09/R10 explicitly report no executed DCC simulation/render/round-trip. Preserve that boundary even where the report includes example hashes and executed mathematical geometry checks.

R11 contains a downloadable 73,202-byte ZIP with 33 members. Its SHA-256 matches the value printed in the report and its archive CRC check passes. We inspected the supplied result records and independently recounted the two OBJ/GLB assets; the counts and opposite-winding relationships match. We did not execute the bundled scripts or repeat all 17 reported checks. The evidence review preserves member hashes, reported results and our narrower observations.

R12 was read in full. Eleven selected files in the local Jolt dependency match public source at the exact pinned commit after line-ending normalization. The bare CPU rod contact gap, root-mass coupling and ignored-contact error bits are supported by inspected code; their practical effect still needs controlled runs. R12 could not audit the local Lab wrapper, and our scoped checks do not qualify that wrapper's caching, scheduling, Undo or publication. The source review records hashes, observations and limits. None of R12-E1 through R12-E8 was executed here.

R13 was read in full, including its harness and source excerpts. Its embedded 92,941-byte ZIP has ten members and passes CRC; all six artifact hashes declared in its provenance agree. The separate 2,307-byte result download is identical to the final archive member. All 48 cited file hashes match the local dependency after line-ending normalization, and all 82 cited ranges exist. We inspected the stored initial/follow-up results and code without building or running them. The evidence review records those checks and the reported values.

R13's execution scope is one Linux/GCC, double-world-position, single-threaded CPU build with compute disabled and no timing measurements. The clamped one-segment rod demonstrates contact coverage, not a fair feasible-routing comparison. The tiny-scale failures have not been traced to a specific rejecting branch; the final rigid penetration is a direct-query residual, not a rendered-surface measurement or universal error bound. Root springs, a complete fitter, custom rod segment correction and host output remain unexecuted. Its post-tag comparison and downloaded archive metadata are supplied provenance, not independently repeated comparisons here.

R14 was read in full, including its 94-entry register. Its 362,872-byte evidence ZIP has 109 members and passes CRC; all six declared artifact hashes agree. We inspected its native/analytic code and records, build recipe/logs, provenance and reproduction wrapper without executing them. Its source ledger matches 49 local file hashes and 86 valid ranges. Two exact member copies provide convenient access to native and analytic JSON. The R14 review preserves the detailed checks.

R14 reports seven native records across five families in a single-precision/SSE2/assert CPU profile, plus nine analytic fixtures containing fifteen stored checks. Its 5/50-iteration three-point crossing remains a clamped detection counterexample. Its free-rod ejection is a transient, not a fit. The tiny-size and large-origin entries in this report are analytic, while R13 has separate native query results. Report-local E1-E7 and P/R/C/U/X citation labels are not library experiment/source IDs. No cross-platform identity, throughput ranking or independent repetition follows from agreement.

The reported tag metadata also reconciles a potential count confusion: R14's saved comparison lists 78 changed files in total, including 47 under Jolt/, which is the narrower scope quoted in R13. This was checked within supplied metadata; the remote comparison and source archives were not independently re-fetched here.

Correction register

ID	Earlier claim or risk	Consolidated treatment
C01	Public-main/version assumptions in R03 used as the local baseline	Use the recorded local dirty tree, current README and exact inspected file hashes; distinguish committed HEAD from working files
C02	Automatic fitting necessarily requires artist timeline playback	Houdini documents static Guide Collide and frame-local Vellum internal roll-forward; neither guarantees this product's full criteria
C03	A pure guide wire can universally drive ordinary Point Deform	SideFX explicitly warns against wire models; use a framed curve method or qualified surface/volume cage
C04	All “quasistatic” controls mean the same operation	Separate simulation type, H22 hair orientation solve and Guide Deform shape relaxation
C05	tyFlow 2.010/older CUDA requirements represent current capability	Latest study and checked download/history use 2.101 and CUDA 12.8; record installed build and resolve older-page conflicts
C06	Hair Bind is the generic tyFlow grass solver	It binds to Ornatrix strands; the proposed guide recipe uses Particle Bind/Angle Bind
C07	One fixed point, two collinear points and a fixed patch provide equivalent anchoring	They constrain position, tangent and full geometric base frame differently
C08	Moving a collider in from a clear pose proves fixed-from-start overlap recovery	Preserve the two initialization histories as separate tests; latest study does so explicitly
C09	A collision-free guide proves a collision-free visible blade	Validate the decoded mesh, frame, width and any postprocessing/render geometry changes
C10	Gap width proves global infeasibility, or blade width should be added twice	Keep envelope accounting explicit; a channel bound is local and three-dimensional escape may exist
C11	Jolt GPU hair limitations reject CPU rods, or existing Lab integration proves rods work	Separate engine paths; inspect pinned contact code and run the same driver fixtures
C12	Houdini Compound means arbitrary convex decomposition	The packed-object primitive Compound path and connected convex hull grouping have different meanings
C13	Reused stable IDs imply fits survive density/Relax changes unchanged	Reuse requires compatible full placement/source transforms and dependencies, consistent with F02
C14	UV/material changes can always skip preparation	Only appearance-only edits qualify; channels used for roots, membership or physics can invalidate it
C15	Automatic guide extraction or four-weight DQ binding is broadly solved	Treat as asset-family hypotheses; audit real sources and prove a minimal decoder first
C16	Research work counts, memory scenarios and example inspection are benchmarks	Keep task groups, arithmetic, document inspection and actual execution separate
C17	Live may universally wait only for mouse release or a frozen cache update	Preserve relevant dependency/animation semantics; a sampled frozen mode needs an explicit new policy
C18	Render-vertex components identify blades or coincident positions prove shared identity	R11's public meshes and synthetic examples distinguish render corners, logical positions and semantic parts; independent coincident pieces must survive

ID	Earlier claim or risk	Consolidated treatment
C19	Opposite-winding duplicate faces necessarily mean corrupt branching topology	Classify explicit two-sided construction; any analysis simplification stays reversible and leaves render data intact
C20	One canonical guide evaluator automatically reproduces arbitrary Skin/cage behavior	Treat R11's mapping as a candidate with supplied bindings; require conversion and stress-pose parity for each supported authoring route
C21	A framed mapping preserves all width, volume, surface area or authored shading	Its single-guide cross-section property is conditional; multi-guide/hold blends can collapse, and authored normal transport is a separate policy
C22	No offline source-preparation evidence is available	R11 adds supplied topology/arithmetic results and independently recountable assets; these do not qualify automatic extraction, private assets or host output
C23	A likely renderer or an SDK normal container establishes delivery support	Select an explicit renderer/build/mode and test mesh, exact-variant, normal and lifecycle paths independently
C24	CPU rod constraints imply whole-segment environmental collision	The checked bare-rod path supplies sampled vertices and one selected plane per vertex; compare actual segment/envelope contact, with subdivision treated as approximation
C25	Zero root inverse mass automatically provides a free hinge	Normal rod initialization couples rod inverse mass to its endpoints and angular corrections; inspect and measure each attachment construction
C26	Closest triangle/backface thickness guarantees recovery from inside a solid	Keep explicit closed-solid containment, open-surface and protected-root policies; finite backward thickness is not a global classifier
C27	A 5.6.1 header or release label identifies the tested engine	Use exact commit, local patches, configuration and binary/loaded hashes; our eleven-file comparison is source-only evidence
C28	Engine reuse automatically supplies wrapper lifecycle and successful completion	Audit cache ownership, bounded cancellation and job/barrier integration; capacity error bits can indicate ignored contacts and must prevent false acceptance
C29	The bare-rod contact finding describes every Jolt soft-body subsystem	Scope it to the inspected rod environmental path; rigid CCD has separate soft-body face handling, and GPU hair is another subsystem
C30	R12's proposed 10,000-body test fits the current Lab profile	Current local limits declare 5,000 bodies; a larger standalone harness needs separate admission/resource scope and changes no current acceptance
C31	A supplied harness/result JSON means we reproduced the run or passed a Cyrus case	R13 reports external headless execution; package/source consistency was checked here, but no engine run or Cyrus acceptance row was completed
C32	No surface hit, or one raw mesh-parity result, proves a blade outside a solid	Test containment independently, including shared-edge/vertex degeneracy; keep an ambiguous status and explicit open-surface policy
C33	Scaling collision tolerance fixes every tiny-geometry failure	R13's two tiny-scale misses persist after that change; qualify numerical normalization and dynamic rescaling separately without inventing a universal minimum size

ID	Earlier claim or risk	Consolidated treatment
C34	Sleeping with zero update errors guarantees penetration below configured slop	R13 records about 4.289 mm query penetration under 1 mm slop; final visible/proxy acceptance remains independent
C35	One whole-segment overlap query establishes continuous bending clearance or universal correction direction	Declare temporal coverage, query shape ordering, penetration/separation sign, COM and offset conventions explicitly
C36	Inverse mass zero, hard skinning or resetting points establishes complete stationary rod state	Initialize root velocity deliberately and reset positions, velocities, rod frames/angular velocities and rest state through a supported whole-body path
C37	More soft-body iterations refresh collision planes at every projection	The checked path reuses planes across soft substeps; world collision steps/manual updates are separate boundaries with one stepping owner
C38	Solver array indices and generated transported frames are an authored blade binding	Preserve semantic remaps through endpoint reversal/optimization, authored material frames and consistent rest-relative rotations
C39	StaticCompound means a static body, or every triangle mesh is strictly static-only	Compound child eligibility and the documented dynamic-mesh mass override matter; mesh-mesh/heightfield restrictions remain, with no automatic Cyrus support extension
C40	R13 and R14 use one build profile, or all their small/large-scale rows are native runs	Preserve R13 double/AVX and R14 single/SSE2/assert receipts; distinguish R14 analytic expectations from R13 native query diagnostics
C41	The public rigid contact callback inserts CPU rod segment constraints	Soft bodies are diverted to a separate pipeline; the soft-body listener exposes pair controls/const observations, not endpoint/frame constraint rows
C42	Per-query locking plus a final revision check makes an iterative fit coherent	Pin one obstacle shape/pose generation for the entire job; lifetime safety and stale-result rejection do not repair mixed-input iterations
C43	A raw overlap witness or rod quaternion directly defines a capsule contact response	Convert rod-Z/capsule-Y frames and qualify material fraction, normal and endpoint/frame gradients; frozen-witness math is only a local model
C44	Dynamic escape from inside a primitive is an accepted static fit	R14's free rod exits with continuing velocity; require mode-specific root/rest/length, motion and final-surface acceptance
C45	Compute hair always requires a scalp authoring stage	The official source includes a no-scalp example; compute integration and sampled collision limits still apply, with no runtime qualification here
C46	A second PhysicsSystem has independent global services or permits arbitrary cross-module shape sharing	Shared-runtime factory/dispatch/allocation hooks need one lifetime owner; inspect actual build/ABI/module boundaries before sharing resources

Primary support: Guide Collide, Vellum Solver, Point Deform, H22 changes, tyFlow download/history, Hair Bind, packed RBD shape controls, and local foundation. The primary reference register records narrower applicability and review status.

Remaining choices

ID	Question	Best evidence to obtain
O01	Which real assets should the first feature support?	Representative production-library audit; R11's two stylized public assets do not measure private-library suitability or preparation time
O02	Which output path faithfully carries distinct bends?	One authored deformation through Max 2027 viewport, selected renderer and Bake
O03	Which fitting method handles fixed-obstacle initial overlap acceptably?	Shared one/two-stone fixtures comparing simple fitting and a rod/constraint route
O04	Is a conservative guide envelope sufficient for the target close-up quality?	Driver-versus-visible-mesh contacts, narrow gaps and twisting ribbons
O05	What root attachment and failure policy feels useful?	Artist review of hinge/soft/fixed examples and explicitly flagged exceptions
O06	What scale and update budget are practical?	Whole-pipeline cold/warm/idle timing, actual interaction and peak/retained memory
O07	How should fit data persist, freeze and participate in Undo?	Scoped ownership/time/persistence design and lifecycle tests
O08	Which inspected engine capabilities should be reused through the local wrapper?	R12 resolves specific dependency paths; still measure cooking/cache, scheduling/cancellation and stable-result integration within the admitted profile
O09	Can the latest recipes be reproduced exactly?	Missing fixture JSON, example manifests and runnable .hip/.max scenes with recorded versions
O10	Does the framed-offset candidate preserve the intended source poses and shading?	Rest reconstruction, twist/frame alignment, junction/blend limits, imported-rig pose comparisons and selected-renderer normal tests
O11	Should the focused fitter query Jolt shapes or a direct triangle BVH?	Same-input whole-segment/multi-contact comparison including cold cooking, warm total cost, memory and lifecycle
O12	Which root construction gives the intended hinge, soft angle or clamp?	Controlled root/rod-mass inspection and measured position, frame and rest-shape response
O13	What numerical scale and solid/surface profile can the first query adapter support?	Independent tiny-crossing, center/off-axis parity, contact-sign/COM and mapped-back normalization controls before admitting production assets

These are experiment priorities. They are not all questions the user must answer before the library can be delivered.

Missing artifacts and future submissions

R09/R10 reference a companion initialization JSON under two filenames, an example-download manifest and extracted archive manifests. Those artifacts were not supplied. R11's separate source-preparation ZIP is now preserved and inspected; it does not supply that earlier grass-initialization JSON. No runnable .hip/.max scene, application execution receipt, representative production-library audit or Cyrus output prototype was included.

The supplied collection is complete through the user's final R14 delivery: fourteen original files, R11/R13/R14 archives and accompanying result copies are preserved. R12 supplied procedures; R13/R14 supply different

reported headless evidence. None supplies the missing R09/R10 artifacts, runnable DCC scenes or a Cyrus execution receipt. Those limitations remain visible without treating the documentation task as unfinished implementation.

Freshness and repository scope

The local knowledge service reported stale evidence when full source verification was requested. Direct source inspection was therefore used for consequential baseline claims. No corpus refresh or activation was performed. The recorded HEAD alone does not identify the dirty working files; use their hashes from the snapshot.

This organization changes research documentation and a navigation link only. It does not modify plugin behavior, publish a website, claim renderer qualification or resume a roadmap. Existing receipts retain their original version, host and applicability limits.

Asset audit worksheet

Template only. No assets have been scored or qualified. Copy the record below per source. Use real observations; leave unknowns explicit.

Field	Record
Asset ID / file / source revision	Pending
Geometry and material hashes	Pending
Host and source-access method	Pending
Geometry family	Clean ribbon / disconnected blades / joined base / branching / scan / opacity card / proxy / other
Vertex and face count / map channels	Pending
Physical size / coordinate frame / transform	Pending
Connected components versus intended blades	Pending
Geometric / render-stream / logical vertex counts	Pending
Verified duplicate correspondence versus independent coincidence	Pending
Opposite-winding pairs, sidedness and analysis-only simplification	Pending
Root identification and ambiguity	Pending
Base tangent and width/frame direction	Pending

Field	Record
Shared-base motion policy and child attachments	Pending
Available longitudinal segmentation	Pending
Driver tier proposed	A clump / B blade or ribbon / C authored advanced
Guide/driver construction method	Pending
Binding and cross-piece restrictions	Pending
Fixed rest coordinates, full residual offsets and unwrapped twist	Pending
Rest reconstruction error and bend/twist/blend stress limits	Pending
Imported rig/cage conversion and pose comparisons	Pending / not applicable
Unbound vertices / cross-binding	Pending
Full canonical proxy geometry available	Pending / not applicable
UV/material channels used as preparation metadata	Pending
Explicit normals / smoothing policy	Pending
Renderer ClassID/build/mode, adapter and delivery qualification	Pending; separate from preparation readiness
Supported scale, mirror and shear policy	Pending
Manual preparation tasks and measured minutes	Pending
Root/frame/contact/output errors	Pending
Rejection or repair cause	Pending / none observed
Screenshots, source and prepared artifact links	Pending
Outcome	Unaudited / needs preparation / experiment-ready / rejected for stated scope

Suggested sample selection

Use multiple geometry families and a realistic range of source complexity. Include at least one intentionally unsuitable source so failure reporting is evaluated. Do not infer a support percentage from a conveniently chosen small sample; record the selection method and intended population.

“Experiment-ready” means the inputs are sufficiently explicit for a proposed test. It does not establish a shipped feature or renderer qualification.

Validation matrix

All rows Pending. These are proposed acceptance experiments, not results. Set physical tolerances, exact versions and budgets in an experiment record before execution. VG references point to the proposed engineering guidelines.

ID	Case	Required observation	Guide	Status
V01	One authored bent blade through viewport/render/Bake	Same intended visible shape, bounds and root	VG12	Pending
V02	Untouched source before/after fitting	No source topology/position mutation	VG02	Pending
V03	Position-only anchor	Root stays fixed while hinge motion is permitted	VG03	Pending
V04	Fixed tangent	First direction respects its declared clamp tolerance	VG03	Pending
V05	Full base frame	Width/roll attachment is measured independently	VG03	Pending
V06	Soft angular base	Root stays fixed; angular response follows selected policy	VG03	Pending
V07	Fixed stone with intersecting rest source	Recovery attempt judged on contact, strain and work limit	VG06	Pending
V08	Clear initial shape with separate rest target	Restoring load and orientation initialization are explicit	VG06	Pending
V09	Moving-stone conditioning	Loading history recorded; not substituted for V07	VG06	Pending
V10	Segment endpoints outside but interior crossing	Whole-segment contact detects obstruction	VG05	Pending
V11	Component entirely inside closed obstacle	Classification catches embedded geometry	VG05	Pending
V12	Guide clear, ribbon edge intersecting	Visible-mesh validation rejects false success	VG05	Pending
V13	Two stones, joint contact	Both obstacles respected in one solve	VG05	Pending
V14	Narrow channel with 3D escape route	Route recorded; no unsupported infeasibility claim	VG05	Pending

ID	Case	Required observation	Guide	Status
V15	Protected root inside obstacle	Explicit root conflict; no movement/deletion	VG01	Pending
V16	Sharp corner and thin obstacle	Proxy/discretization limitations diagnosed	VG05	Pending
V17	Open or invalid obstacle mesh	Declared sign/contact policy enforced	VG04	Pending
V18	Two neighboring independent blades	No cross-binding or accidental joint identity	VG02	Pending
V19	Joined-base and opacity-card sources	Correct tier or explicit preparation limitation	VG02	Pending
V20	Nonuniform, mirrored or sheared transform	Supported evaluation or clear rejection	VG12	Pending
V21	Stone moves into/out of contact	New and released plants reconsidered	VG07	Pending
V22	Stone crosses field or is deleted	Old/new regions and prior dependencies included	VG07	Pending
V23	Same inputs, fresh versus repeated update	Declared repeatability and rest policy hold	VG06	Pending
V24	Density/quota/Relax or source swap	Full-input compatibility governs reuse	VG08	Pending
V25	Appearance versus semantic channel edits	Only compatible appearance changes reuse shape	VG08	Pending
V26	Manual pending edit	No refit before Update; prior output marked stale	VG11	Pending
V27	Relevant Live edit/animation	Correct dependencies schedule bounded updates	VG11	Pending
V28	Navigation/selection/unrelated animation	No static solve or unchanged buffer upload	VG11	Pending
V29	Cancellation and stale worker completion	No late or partial publication	VG09	Pending
V30	Owner deletion, reset and shutdown	Workers/readers retire safely	VG09	Pending
V31	Allocation/work-limit failure	Previous complete publication retained	VG10	Pending
V32	Undo/Redo and repeated history	Coherent state restored; memory release measured	VG13	Pending
V33	Save/reopen and missing dependencies	Current/stale/unavailable states preserved honestly	VG14	Pending
V34	UV seams, material IDs and authored normals	Appearance policy matches across output paths	VG15	Pending
V35	Smooth/shell/subdivision output changes	Final evaluated contact revalidated or unsupported	VG12	Pending

ID	Case	Required observation	Guide	Status
V36	Many unaffected placements, sparse local contact	Work and output scale with relevant changes	VG13	Pending
V37	Dense contact/high-vertex sources	Whole-pipeline peak/work budgets measured	VG13	Pending
V38	Approximate deformed variant reuse	Error budget, contact and popping revalidated	VG12	Pending
V39	Optional flagged partial acceptance	Complete atomic generation with explicit exception policy	VG14	Pending
V40	Rigid versus deformable leaf	Mode-specific expected behavior remains distinct	VG16	Pending
V41	Render/logical/semantic identity and independent coincidence	Verified duplicate mapping preserves seams without welding independent pieces	VG17	Pending
V42	Explicit opposite-winding faces	Analysis classifies sidedness while original render faces/channels remain intact	VG17	Pending
V43	Curved rest binding and complete-turn twist	Rest reconstruction, fixed coordinates and frame/twist sampling meet declared limits	VG18	Pending
V44	Opposing-guide and soft-hold blends	Width/area collapse is detected; limited preservation claims remain conditional	VG18	Pending
V45	Shared vertex with competing blade motion	One explicit shared-base/attachment policy resolves or rejects the request	VG18	Pending
V46	Imported bone/cage preparation	Conversion passes representative pose comparisons or remains unsupported	VG18	Pending
V47	Borrowed/shared mesh channels and renderer caches	Owned adapters survive actual readers without source mutation or early release	VG19	Pending
V48	Object offsets, parents and world-space modifiers	Evaluated source and transform are captured coherently and applied once	VG19	Pending
V49	Mirrored Bake with corner channels	Winding, map/normal indices and tangent handedness stay consistent	VG15	Pending
V50	Non-geometric authored foliage normals	Selected renderer/custom mesh/Bake paths honor the declared normal policy	VG19	Pending
V51	Exact solver source/build/loaded identity	Commit, local patches, build flags, precision, enabled modules and binary hashes agree with the inspected baseline	VG20	Pending
V52	CPU rod root-mass coupling	Initialized rod inverse mass, canonical zero root velocity and measured angular behavior distinguish position-only, soft and hard attachment	VG03	Pending
V53	Direct BVH versus Jolt contact queries	Same fixtures expose contact agreement, multi-contact, cold/warm cost, storage and lifecycle differences	VG20	Pending

ID	Case	Required observation	Guide	Status
V54	Engine contact/pair/manifold capacity exhaustion	Error bits prevent false acceptance; failed successor preserves the previous complete publication	VG21	Pending
V55	Scene units, scale and distant origin	Equivalent physical inputs and local-origin transforms preserve declared contact/root quality within measured tolerances	VG21	Pending
V56	Engine scheduling and cooperative cancellation	Bounded update boundaries, nested jobs/barriers, shutdown and measured cancellation latency satisfy the adapter contract	VG21	Pending
V57	Closed-mesh parity degeneracy	Center/off-axis, shared-edge/vertex and open-mesh controls produce verified solid classification or an explicit ambiguous state	VG22	Pending
V58	Tiny-geometry contact regression	Equivalent crossings, scaled tolerance and proposed normalization are checked against analytic overlap and mapped-back units	VG22	Pending
V59	Shape-query construction and coordinates	Capsule-Y/rod-Z conversion, dimensions/scale, query sign, separation depth, base offset and COM/body-origin transforms match known primitive controls	VG22	Pending
V60	Canonical rod rest/reset/remaps	Particle and rod velocities/frames reset coherently; authored rest state and semantic mapping survive property calculation and optimization	VG23	Pending
V61	Soft iterations versus contact refresh	Timing/substep changes are distinguished from collision-plane refresh, with one stepping owner and no double update	VG23	Pending
V62	Sleeping rigid body with residual overlap	Independent final contact/visible-proxy metrics govern acceptance even when sleep and zero error bits are reported	VG22	Pending
V63	Static pose versus deformation trajectory	Declared final-pose or continuous-path contract is tested; a rigid translation cast does not certify arbitrary bending motion	VG05	Pending
V64	Independent oracle and equal-contract comparison	Separate checks expose shared false negatives; both candidates include the contact adapter and accept feasible companions as well as rejecting invalid inputs	VG22	Pending
V65	Immutable query generation during obstacle edits	Every iteration uses one owned shape/pose generation; per-query locking and a final revision check cannot substitute for coherence	VG24	Pending
V66	Shared runtime and multiple solve worlds	Actual module/build/allocator ownership supports service overlap and shutdown without replacing global hooks or retiring another world's resources	VG24	Pending
V67	Qualified witness and endpoint/frame response	Material fraction, normal, mobility weights, pinned endpoints and zero-denominator conflicts pass frozen-witness controls before full nonlinear fitting	VG25	Pending

ID	Case	Required observation	Guide	Status
V68	Dynamic interior ejection versus accepted static fit	Contact and escape alone do not pass; final velocity, root/rest/length and visible clearance meet the selected mode's stopping criteria	VG06	Pending

Required result fields

Store pass/fail/pending per row, numeric errors with units, declared tolerances, accepted frame or work count, source/obstacle identities, output artifacts and any unsupported scope. Include a reason for skipped rows. A documentation integrity pass does not change this matrix's Pending status.

Solver decision experiments

Eight proposed experiments from R12; all Pending. This is a focused decision checklist within the validation plan, not an implementation authorization or a second progress ledger. Use an experiment record and preserve failures alongside successes. Source review is recorded separately in R12-source-review.json.

Sequence and shared controls

Confirm identity, then run tiny root/contact probes alongside the first authored Max 2027 viewport/render/Bake experiment. R12 proposes native fixtures before UI work; the consolidated plan keeps output feasibility early as an independent risk. Neither requires a substantial production solver first.

Use the same prepared geometry, units, rest state, initial pose, root policy, decoder and contact tolerances across methods. Record timestep, iterations, physical scale, local origin, supported transforms and budgets. V55 unit/translation invariance and V56 scheduling/cancellation checks apply across the engine experiments. Every status below remains Pending; inspected source is not a fixture pass.

R13 evidence and follow-up gates

R13 adds a reproduction archive and reported result JSON. Our review verifies package/source consistency without running the harness. These records partially exercise the questions behind E3/E4/E5/E7 in one external build; they do not complete those Cyrus experiments. In particular, the clamped one-segment rod does not test feasible routing, and the gap rows are queries rather than fits.

All five R13 follow-up gates remain Pending. They refine the shared plan as follows:

Gate	Decisive evidence needed	Shared plan mapping
G0 Geometry adapter	Independent primitive truth, occupancy ambiguity, query sign/COM, tiny-scale normalization and rebasing through actual snapshots	E1/E3/E4/E6; V51, V53, V55, V57-V59, V64
G1 Root/rest/state	Curved and twisted guide, zero initial root velocity, free/soft/hard angle response, semantic remaps and complete reset	E2; V03-V06, V43, V52, V60

Gate	Decisive evidence needed	Shared plan mapping
G2 Contact correction	Feasible multi-segment one/two-stone route, inside start, root conflict and sharp corner with an independent final validator	E3/E4/E5; V07-V17, V61, V63
G3 Equal-contract comparison	Fitter versus rods with the required contact adapter; same evaluator/tolerances before measuring time and memory	E6; V37, V53, V64
G4 Host/result lifecycle	Changed input during work, cancellation, reset/reload, selected-renderer/Bake/picking parity and retirement	E8; V01, V21-V35, V47-V50, V56

A separate rigid acceptance probe must preserve the reported sleep-with-residual counterexample (V62). The reported 66-update result is not a clearance pass, a throughput measurement or completion of E7's population/lifecycle scope.

Final study R14 and identifier mapping

R14's evidence package contains a different native harness/build and separate analytic oracles. Its report-local E1/E2 supply reported witness/containment baselines, not completion of our similarly numbered experiments. Map R14-E1/E2 into our E1/E3/E4 questions; R14-E3 into root experiment E2; R14-E4/E5 into contact/backend experiments E3-E6; R14-E6 into lifecycle/output E8; and R14-E7 into equivalent-acceptance cost work E6/E7. All of our eight statuses remain Pending.

Use the review receipt for exact configuration and evidence boundaries. H02/H02b compare 5/50 soft iterations on a clamped three-point rod; H04b is free-rod ejection, not a static-fit success. Nine analytic fixtures/fifteen reported checks establish expectations, not solver passes. Add feasible companions, one coherent query generation (V65), shared runtime lifetime (V66), qualified witness/endpoint response (V67) and an ejection/stopping control (V68).

R12-E1: exact-pin build and provenance

Status: Pending. Matrix: V51. In a later authorized task, build the chosen harness against commit `1b2e77e3c6c09709e5edc79dc208459b8ff32051` and record local patches, compiler/configuration, precision and enabled modules, job adapter and binary hashes. Record loaded identity when entering Max; header version 5.6.1 alone is insufficient.

Decision: do not attribute an observed result to the inspected source until the built/loaded identity is established. Our eleven-file source comparison is a useful prerequisite, not completion of this experiment.

R12-E2: root anchoring modes

Status: Pending. Matrix: V03-V06, V52. Use one curved eight-segment rod without obstacles. Compare zero-inverse-mass root initialization with an explicitly designed position attachment and a reference-rod construction. Treat those alternatives as prototypes whose constraints need inspection, not ready-made equivalent controls.

Measure root drift, first-segment angle, material-frame angle, length and rest-curvature error under the same small load. Check the adjacent rod's initialized inverse mass and effective angular response. R14 clarifies that the reference rod can be a separate fixed pair, excluded from visible binding; preserve its remap and never use a

zero-length dummy. **Decision:** identify separate position-only, soft angular and hard angular policies, or reject a construction that cannot provide the intended mode.

R12-E3: segment-crossing falsification

Status: Pending. Matrix: V10, V12. Place endpoints outside a sphere/box with the segment interior crossing it. Compare stock rods, the focused fitter and its contact backends; repeat with 1, 2, 4 and 8 subdivisions and a declared radius/visible-width envelope.

Measure generated contacts and independently computed residual segment/capsule and decoded-mesh penetration. **Decision:** a missed crossing rejects that configuration as a complete contact solution. Improvement with subdivision is an approximation result, not proof for arbitrary obstacle feature sizes or continuous motion.

R12-E4: deep-inside and protected-root cases

Status: Pending. Matrix: V11, V15, V17. Put a movable guide entirely inside a convex solid, then inside a closed triangle mesh; repeat with a protected root inside each. Include an open surface as a separately declared policy.

Measure containment classification, available contacts, exit direction, strain and final diagnostic status. **Decision:** require explicit root-conflict and solid/surface preflight regardless of solver. Preserve roots and plants; finite triangle backface thickness is not a global inside test.

R12-E5: two obstacles, narrow gap and sharp corner

Status: Pending. Matrix: V13, V14, V16. Test two stones together, a channel above/equal/below the declared effective diameter, and a right-angle corner. Retain opposing contacts separately. Add a possible out-of-plane escape route and a visible ribbon whose width changes the available route.

Measure simultaneous contacts, oscillation, penetration, root/frame error and strain under a fixed work limit. **Decision:** select the method that meets those criteria or reports an honest failure. A blocked specified channel does not prove every three-dimensional route infeasible.

R12-E6: Jolt capsule queries versus direct BVH

Status: Pending. Matrix: V10-V17, V53. Use identical segment capsules against analytic, convex, compound and mesh obstacles with matched geometry/representation tolerances. Keep the fitter unchanged when switching contact backends, and use an independent reference where practical to identify shared false negatives.

Measure contact agreement and multi-contact completeness, cold cooking, warm query/update time, peak/retained storage, cancellation latency and lifecycle cost. Keep query-level and complete-pipeline timings separate. **Decision:** reuse Jolt queries only if they reduce total implementation/maintenance risk within the measured quality and resource budgets; otherwise retain direct BVH contact. No winner is established yet.

R12-E7: separate rigid-debris baseline

Status: Pending. Matrix: V37, V40, V54-V56. R12 proposes 100, 1,000 and 10,000 dynamic convex/compound stones over fixed irregular terrain. The current local Lab declares a 5,000-body default/admitted profile. Keep experiments within that profile; the 10,000-body proposal needs a separately scoped standalone harness and admission/resource decision. This checklist raises no existing limit.

Measure residual overlaps, settling criteria, update error bits, peak memory, repeatability and completed stable-ID transform capture. Add deliberately bounded capacity-error cases. **Decision:** qualify rigid settling

independently; sleep alone or a normal update return cannot establish accepted quality or qualify vegetation bending.

R12-E8: shape parity and atomic publication

Status: Pending. Matrix: V01, V02, V29-V35, V47-V50. Start with one authored bend through the minimal evaluator; then use two placements of one prepared source bent differently around two stones. Record accepted placement IDs, immutable source data, completed bounds and exact selected renderer/build/mode.

Compare visible positions, root/frame registration, UVs/materials/normals and Bake output. Exercise cancelled, stale and failed successors while readers use the previous complete result. **Decision:** stop expansion if distinct deformed output or coherent publication fails. Guide-level contact success cannot bypass this gate.

What to bring back

Attach exact fixtures, source/build/binary identity, settings and tolerances, per-case measurements, work/memory/cancellation records, diagnostics and output artifacts. Mark unsupported or skipped cases explicitly. A later choice may adopt a method only for the scope actually demonstrated; no reported estimate or source inspection changes a Pending row to Pass.

Corrected recipe planning matrix

Documentation-derived proposals; no application executions. Recipe IDs belong to the latest supplied R09/R10 study. Its numeric presets and task groups are starting procedures, not measured convergence or efficiency.

Recipe	Inputs / purpose	Evaluation	Main unresolved acceptance
H1B	One guide, final stone SDF, root mask, named mesh	Static geometry	Hard roots, length, mesh/frame clearance
H2B	Same with both stone fields together	Static geometry	Joint contact and preserved narrow gap
H1C	Straight rest constraints, clear initial guide, fixed stone	Frame-local internal settle or dynamic	Fresh orientations, root target, visible mesh
H2C	Three named guides, two fixed stones	Frame-local internal settle or dynamic	3D frame/twist decoding and both-side mesh contact
H1A	Straight source, stone moving in then held	Dynamic conditioning	Loading history, root clamp and settling
H2A	Two moving stones then held	Dynamic conditioning	Approach-order dependence and joint contacts
H3	Packed pebbles, clear starts, static terrain	Rigid timeline simulation	Hull/terrain fidelity and accepted rest state
H4R	Thin rigid leaf proxies at curb	Rigid timeline simulation	Proxy silhouette, padding and leaning pose

Recipe	Inputs / purpose	Evaluation	Main unresolved acceptance
H4D	Segmented unrooted leaf surface	Cloth timeline simulation	Thickness, self/contact, output surface
C	Clear spline, fixed roots/tangent, flat-angle target, fixed stone	tyFlow binding timeline simulation	Frame convention, segment collision, visible blade
C2	Three guides, two fixed stones	tyFlow binding timeline simulation	Correspondence, simultaneous contacts and route
A	Root-tagged ribbon, moving stone then hold	tyFlow Cloth Bind / CCCS	Noncollinear base patch, initial clearance, output
B	Three ribbons, two moving stones then hold	tyFlow Cloth Bind / CCCS	Conditioning history and joint geometry
TF-P	Pebbles and fixed terrain, convex dynamic hulls	tyFlow PhysX rigid simulation	Static mesh terrain, gravity units, cache output
TF-LR	Thin rigid leaves, ground and curb	tyFlow PhysX rigid simulation	Curled silhouette approximation and contact offsets
TF-LD	Unrooted segmented leaves, clean curb/floor	tyFlow Cloth Bind / CCCS	Initial intersections, generated cloth mesh and cache

The earlier supplied matrix R05 has 12 rows and 20 columns. It remains unchanged under originals. This table deliberately uses the later study's more explicit initialization/evidence distinctions; it does not silently rewrite R05's historical content.

The fixed-initialization guide branches reference a missing companion JSON. Before exact reproduction, obtain it or label a reconstructed fixture as newly derived. Optional detailed mesh bindings add preparation and final-surface validation beyond the core driver recipe.

Original research register

Fourteen supplied files, including the final R14 delivery, preserved intact on 24 October 2026. Research content is source material, not instructions to execute. The originals remain in Downloads; this folder contains verified copies.

ID	Research	Original	Relationship / caution
R01	Rooted vegetation engineering report	HTML	Initial engineering study
R02	Vegetation contact report	HTML	Initial contact study
R03	Vegetation interaction engineering report	HTML	Public-repository baseline requires correction
R04	Extended vegetation contact report	HTML	Separate expanded report; preserve original (1) identity
R05	Earlier recipe matrix	CSV	Companion to R06/R07; several claims corrected in consolidated docs
R06	Earlier workflow study, Markdown	MD	Paired narrative with R07; contextualized by later R09/R10
R07	Earlier workflow study, HTML	HTML	Presentation counterpart to R06
R08	Source preparation and geometry delivery	HTML	Asset preparation and output hypotheses
R09	Practical vegetation workflows, HTML	HTML	Latest supplied workflow study; primary recipe reading copy
R10	Practical vegetation workflows, PDF	PDF	64-page presentation of R09 study; not byte/text identical
R11	Source preparation and geometry delivery, evidence-backed revision	HTML	Extends R08; includes two public sources, synthetic fixtures and supplied arithmetic/topology evidence
R12	Collision and solver reuse implementation study	HTML	Exact-pin Jolt path comparison, contact/root gaps and eight proposed experiments; no runtime tests
R13	Jolt collision and solver reuse, with headless probes	HTML	Extends R12 with an authored harness, reported Linux CPU diagnostics, source ledger and G0-G4 follow-up gates
R14	Final collision and solver reuse study	HTML	Distinct single-precision native harness, analytic fixtures, callback/ownership details and report-local E1-E7 gates; supplements R13

Provenance and reading order

The manifest records original absolute paths, byte counts, SHA-256 hashes and relationships. All fourteen destination hashes matched their source files when copied. R06/R07 are paired earlier narrative formats; R09/R10 are the later practical study in HTML and PDF. Related formats are not independent experiments. The latest original PDF has 64 pages.

Start with the consolidated chapters, then R14 for the final implementation-boundary study, R13 for its different diagnostic profile, R12 for source-only framing, R09/R10 for detailed vendor recipes, R11 for preparation, R08 for

its earlier study, and R01/R02/R04 for engineering depth. Read R03 and R05-R07 with the correction register. All original claims remain unchanged in these files.

R11 embedded evidence

The exact evidence ZIP was decoded from R11, preserved separately and checked against its printed SHA-256. It contains 33 files: the inspected public OBJ/GLB/MTL assets and license, seven synthetic fixture pairs, scripts, figures, and reported audit/evaluator results. The review receipt records every member hash and our independent two-asset topology recount. The bundled scripts were archived without execution; their stored 17 results are reported evidence. This archive does not replace the missing R09/R10 fixture JSON.

R13 embedded reproduction package

The exact ZIP and final result JSON were decoded from R13 without executing page scripts. The 92,941-byte ZIP has ten members: authored C++/CMake, instructions, initial/final results, provenance, source index/excerpts, changed-file list and license. The separate 2,307-byte JSON is byte-identical to its final result member. CRC and all six declared artifact hashes pass. The review receipt also records 48 matching local source-file hashes and 82 valid cited ranges.

The reported run used one Linux/GCC CPU configuration; no timings, full fitter, Max/Cyrus or GPU-hair execution was supplied. The upstream Jolt archive and compiled probe binary are not bundled. Reproduction commands are preserved as source material, not commands run for this documentation update.

R14 embedded evidence

R14's 109-member archive preserves its code, logs, provenance, source excerpts and license. Exact member copies expose seven native records and nine analytic fixtures/fifteen stored checks. These two JSON files are convenience copies, not additional research submissions. The review receipt records CRC, six declared member hashes, 49 matching local source files and 86 valid ranges. Bundled commands were not run here. The source register's P/R/C/U/X labels and E1-E7 gates are local to that report.

Dependencies that were not supplied

R12's source study remains preserved with its source review. R13 now adds a separate headless seed; it does not fill the earlier vendor/DCC artifact gaps below.

- common_grass_initialization.json, also called CyrusScatter_Common_Grass_Initialization.json.
- The R09/R10 author's example_downloads.json and extracted archive manifests.
- Runnable Houdini .hip or Max/tyFlow .max scene fixtures and execution receipts.

These were not recreated or presented as original evidence. The provided reports remain useful without them, but exact reproduction needs those dependencies or a clearly labeled newly derived fixture.

Adding a later submission

Copy it under a new source ID and preserve the exact filename and bytes. Record its relationship to prior reports, review substantive changes, and update the correction register. Retain earlier originals and their hashes.

Primary reference register

Review date: 24 October 2026. “Checked” means documentation or source was inspected during the research review conversation, including the current organization. It does not mean the application workflow was executed. This is the selected reference set supporting the consolidated documents; the originals retain their larger bibliographies.

ID	Primary source	Supports / limitation
P01	SideFX Guide Collide With VDB	Checked: static curve/SDF operation; no full product root/mesh guarantee
P02	SideFX Vellum Solver	Checked: Simulation Type/internal roll-forward, target and thickness semantics
P03	SideFX H22 Vellum changes	Checked: orientation quasistatic and damping are distinct H22 changes
P04	SideFX Guide Deform	Checked: current guide deformation/shape relaxation; no documented obstacle-contact input
P05	SideFX Point Deform	Checked: wire-model warning and piece restrictions
P06	SideFX Wire Deform	Checked: correspondence and frame-aware curve decoding
P07	SideFX Path Deform	Documentation-based decoder proposal in R09; current parameter assembly still needs local validation
P08	SideFX Vellum Constraints	Root/orientation and geometry/constraint stream documentation cited by R09; no complete scene execution
P09	SideFX packed RBD object	Checked: Compound primitives differ from connected convex-hull grouping
P10	tyFlow release history	Checked: 2.101, CUDA 12.8 history and version-specific corrections
P11	tyFlow download	Checked: vendor distribution entrypoint and example archive listing; installed package not inspected
P12	tyFlow system requirements	Checked: older requirements conflict with later release information; solver paths differ
P13	tyFlow CCCS FAQ	Checked: initial intersections, fallback and nondeterminism limits
P14	tyFlow Particle Bind	Checked: bind-segment collisions, radius and same-event object collisions
P15	tyFlow Hair Bind	Checked: Ornatrix attachment and Position Hair data requirement
P16	tyFlow Rotation	Report-backed documented boundary: core bind solver does not solve particle rotation
P17	tyFlow main settings	Report-backed documented history-dependent/independent distinction

ID	Primary source	Supports / limitation
P18	tyFlow reset FAQ	Report-backed reset/cache lifecycle; not a Cyrus behavior contract
P19	tyFlow FREE license	Report-backed vendor feature restrictions; verify chosen license/package before reproduction
P20	tyFlow tyParticleSkin	Checked: reference binding and element matching exceptions
P21	Jolt 5.4.0 release	Checked: CPU Cosserat-rod feature; no Cyrus rooted implementation qualification
P22	Jolt 5.6.0 release	Checked: separate experimental GPU hair path and its scope
P23	Autodesk Max 2027 GeomObject	Checked: render interface context; adapter/lifetimes still require a local design/test
P24	Autodesk Max 2027 render-time instancing	Checked: source/instance output interface; not automatic per-placement deformation
P25	Autodesk Max 2027 ISkinContextData	Checked: bone weights plus per-vertex DQ blend state
P26	Autodesk Max 2027 Mesh	Checked for R11 update: shallow/shared channels, deep copying and renderer-attached cache data; no consumer-lifetime qualification
P27	Autodesk Max 2027 MeshNormalSpec	Checked: explicit corner normals and limited-pipeline warning; actual renderer/stack parity remains a test
P28	SideFX Orientation along Curve	Checked: tangent/up-vector frame construction; no automatic recovery of intended source twist
P29	Kenney Nature Kit	R11 supplies two public sources and original license; this update recounts embedded OBJ/GLB topology, not the complete vendor archive or private library
P30	Autodesk Max 2027 INode	Source for coherent evaluated-mesh/object-offset/world-space transform capture; Max integration still pending
P31	Pinned Jolt Core header	Checked local/public source match: 5.6.1 header; exact commit/build identity remains necessary
P32	Pinned soft-body vertex iterator	Checked: sampled vertex/inverse-mass inputs and selected plane; no rod-segment geometry in this interface
P33	Pinned soft-body shared settings	Checked: rod-property calculation uses minimum endpoint inverse mass; practical attachment response untested
P34	Pinned soft-body motion properties	Checked: environmental vertex contact, rod angular corrections and fixed-vertex projection skip
P35	Pinned triangle contact helper	Checked: finite backward thickness; not arbitrary closed-solid containment
P36	Pinned narrow-phase queries	Checked: shape-query API/base-offset context; performance and fitter integration remain Pending
P37	Pinned update errors	Checked: manifold/pair/contact-capacity error bits can mean ignored contacts

ID	Primary source	Supports / limitation
P38	Pinned GPU hair interface	Checked: separate experimental GPU subsystem; not qualification of CPU rods or Max output
P39	Pinned GPU hair implementation	Checked: narrower collision-shape collection; no current adoption or GPU benchmark
P40	Pinned physics settings	Checked: engine defaults are numerical configuration, not measured artist clearance presets
P41	Pinned soft-body creation settings	Checked: body-wide vertex-radius/control defaults; no variable ribbon-envelope guarantee
P42	Pinned physics-system implementation	Relevant local CCD/soft-body-face path inspected; limits the bare-rod finding rather than proving all contact paths equivalent
P43	Pinned overlap-query results/settings	Checked local source/hash: shape-2 penetration axis, negative separated depth, dimensional collision versus dimensionless penetration tolerance
P44	Pinned capsule shape	Checked with paired implementation: local-Y half-height, explicit zero-length/settings fallback and uniform-absolute-scale requirement
P45	Pinned rod data and optimization remaps	Checked local source/hash: semantic index mapping and rest/material-frame preparation obligations
P46	Pinned soft-body runtime interface	Checked: rod getters, COM-relative state and manual-update warnings; no complete adapter qualification
P47	Pinned mesh point-parity fallback	Checked: raw positive-Y triangle-hit count; R13 supplies a center/off-axis degeneracy record, not our repeated engine run
P48	Pinned architecture documentation	Checked ranges/hash: shape roles, COM, units, callbacks, translational CCD and replay limits; examples are not Cyrus acceptance
P49	Pinned mesh shape contract	Checked: conditional dynamic/kinematic support, supplied mass properties, unsupported mesh pairings and manifold occupancy assumptions
P50	Pinned body reset/transform interface	Checked with implementation: ApplySoftBodyCreationSettings requires a body outside the physics system; origin differs from COM
P51	Pinned hair collision shader	Checked source/hash: separate per-non-root-vertex winning-plane path; no GPU runtime result
P52	Pinned EPA hull builder	Checked source/hash: additional absolute degeneracy threshold; not a traced explanation of R13's tiny-scale misses
P53	Pinned CMake build options	Checked source/hash: 5.6.0 project string alongside 5.6.1 header, separate precision/determinism flags; exact revision remains identity
P54	Pinned soft-body contact listener	Checked: pair settings/acceptance and post-solve observation; no segment endpoint/frame constraint insertion interface

ID	Primary source	Supports / limitation
P55	Pinned soft-body manifold	Checked source/hash: const particle/contact view; callback diagnostics are different from adding solver constraints
P56	Pinned rigid contact-generation hook	Checked with ProcessBodyPair: public rigid hook exists; soft-body pairs are diverted before its invocation
P57	Pinned no-scalp compute-hair sample	Source inspected: scalp data/joints are optional in this example; no hair runtime or host output qualification

Verification limits

Live vendor pages can change. Keep access dates and exact installed versions when executing a recipe. The checked/report-backed distinction above prevents this register from implying a fresh independent audit of every citation in all supplied originals.

The source manifest hashes the user-supplied documents, not remote webpages or vendor example binaries. HDA hashes printed in R09/R10 are the author's reported evidence; the corresponding downloaded-asset manifest was not supplied.

R11 differs: its source-preparation evidence ZIP is embedded in the supplied HTML. We preserved it, checked its printed hash and CRC, inspected its stored results and independently recounted the public mesh topology. The review record distinguishes that work from the report author's 17 checks. The bundle's scripts were not executed here.

For R12, the source review records eleven local/public file comparisons at one exact Jolt commit, including hashes and newline-normalized equality. P42 is an additional locally inspected path, not part of that eleven-file public comparison. These checks establish source behavior only. They do not identify a built/loaded engine or complete any proposed native/Max experiment.

R13's evidence review records all 48 cited full-file hashes matching the local dependency after line-ending normalization, with 82 valid ranges. Its original ZIP retains the larger source index/excerpts and reported build/results. We inspected those data without running the harness; the supplied Linux diagnostics, our source checks and future Cyrus acceptance are separate evidence categories.

R14's review records 49 local full-file matches, 86 valid ranges and a 94-entry report register. Its native/analytic evidence uses different labels and build scope from R13. The exact ZIP retains its build logs, supplied API/tag metadata, source excerpts and license. Those data were inspected without executing the bundled code. P48 also supports the shared-runtime service distinction in its multiple-physics-systems section; actual Cyrus module/build ownership remains a later audit.